Living Stone International School, Inc.
(A Non-stock, non-profit Educational Institution)
43-B Marlim Blvd. Extension Corner Debra Street, Pulung Maragul
Angeles City. The Philippines 2009
Telephone No. (045) 885 2596, (045) 436 2704
Jesus first. Others second. Yourself third.

Qualitative Research

The Influence of Blue Light Exposure on Sleep Quality and Anxiety in STEM Students ranging 15 to 16

Senior High School Department of

Livingstone International School

Written by

Manuel Seraphim

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Abstract

This qualitative study investigates the impact of blue light emitting from digital devices on the sleep quality and mental health of 15- to 16-year-old adolescents in STEM tracks. This study examines adolescents' awareness, behaviors, and coping strategies regarding blue light exposure through semi-structured interviews and open-ended surveys. Modern-day STEM students aged 15 to 16 are among those most affected by extended screen time for both academic and recreational purposes. Blue light disrupts circadian rhythms by suppressing melatonin production, critical for initiating and maintaining sleep. This disruption leads to delayed sleep onset, reduced sleep duration, and poor sleep quality. Poor sleep, in turn, exacerbates mental health challenges, including heightened risks of anxiety, depression, and impaired cognitive performance. Results reveal a significant gap between participants regarding their awareness of blue light's adverse effects and actionable behavioral changes. Participants were aware of coping mechanisms, such as blue light filters and reduced screen brightness; however, they responded with inconsistencies in application to decrease screen time, highlighting the need for comprehensive interventions. This study underlines the importance of addressing blue light exposure, sleep quality, and mental health, advocating for educational programs, parental guidance, and technological tools to mitigate adverse outcomes and protect the younger generation. By tackling these critical issues related to the effects of blue light, this research aims to improve adolescents' overall well-being, academic performance, and emotional stability in an increasingly digital world.

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CHAPTER I

INTRODUCTION

Background of the study

The twenty-first century has a high exposure to blue light from electronic devices such as smartphones, tablets, and computers. This exposure towards adolescents has links to disruption in the Circadian rhythms and sleep quality, with effects on mental health. Adolescents aged 15 to 15 in STEM tracks experience elevated screen time due to demands of education and leisure, increasing their vulnerability to adverse Blue Light effects. Adolescents are susceptible to development shifts in circadian timing, making them prone to delayed sleep phases. Further research indicates evening blue light exposure significantly suppresses melatonin production in a study by Behar and Radrizzani (2022) and Prather and Hurst(2022) highlighting how disrupted sleep patterns can affect mental health issues such as anxiety aid depression.

Statement of the Problem

Despite large evidence and knowledge spread online on the adverse effects of blue light, there remains a lack of comprehensive understanding among adolescents about how their screen habits impact their health. Unaware of these physiological and psychological consequences prolonged evening exposure to blue light despite existing interventions such as blue light filters and recommendations for screen reduction desired to be applied consistently to lower reduction, Addressing these gaps is critical to mitigate the risks associated with blue light exposure and improve adolescents' sleep hygiene and mental health. To know how to respond and help the students this research seeks to answer the following:

1. How do STEM students individuals aged 15 to 16 perceive the impact of blue light exposure on their sleep quality and emotional well-being?
   This is the main question aimed to get an idea of the subjective interpretations and personal experiences related to the issue of blue light exposure and its impact on sleep

quality and emotional well-being during adolescence. Striving to identify whether adolescents are aware that their exposure to blue light influences their circadian rhythm,

sleep quality and duration, and other secondary outcomes on mental health by impacting mood, stress levels, and emotional regulations listed within this study.

1.1. What are the participants' personal experiences regarding the effects of screen time before bed?

1.2. How do participants describe changes in their emotional state related to their screen time habits?

1.3. What coping mechanisms do participants use to manage blue light exposure and its impacts on sleep?

2. (Awareness) How do STEM students aged 15 to 16 perceive the existence of blue light and its impact? What do they believe or think about it and its role in affecting sleep and mental health?
3. (Behavioral Patterns) How do STEM students aged 15 to 16 use blue light-emitting devices such as smartphones, tablets, or computers in the evening? What is their habit or routine that would increase the exposure?
4. (Coping Strategies) What coping strategies do STEM students aged 15 to 16 use to reduce their exposure to blue light, including blue light filters, limiting screen time at night, or participating in other sleep hygiene behaviors?

This investigation will gain an understanding of how youth conceptualize and cope with the advancements in technology use, sleep health, and emotional well-being, which in turn could help highlight the need for educational or technological-bad interventions to address these.

Hypothesis/ses of the Study

1. Increased blue light exposure in the evening is associated with delayed sleep onset and reduced sleep quality in adolescents.
2. Poor sleep quality mediates the relationship between blue light exposure and increased mental health challenges such as anxiety and depression.
3. Adolescents with higher awareness and coping strategies (e.g., using blue-light filters) will report better sleep and mental health outcomes. All of which provide honest and reflective answers during interviews and surveys of their actual habits and strategies.

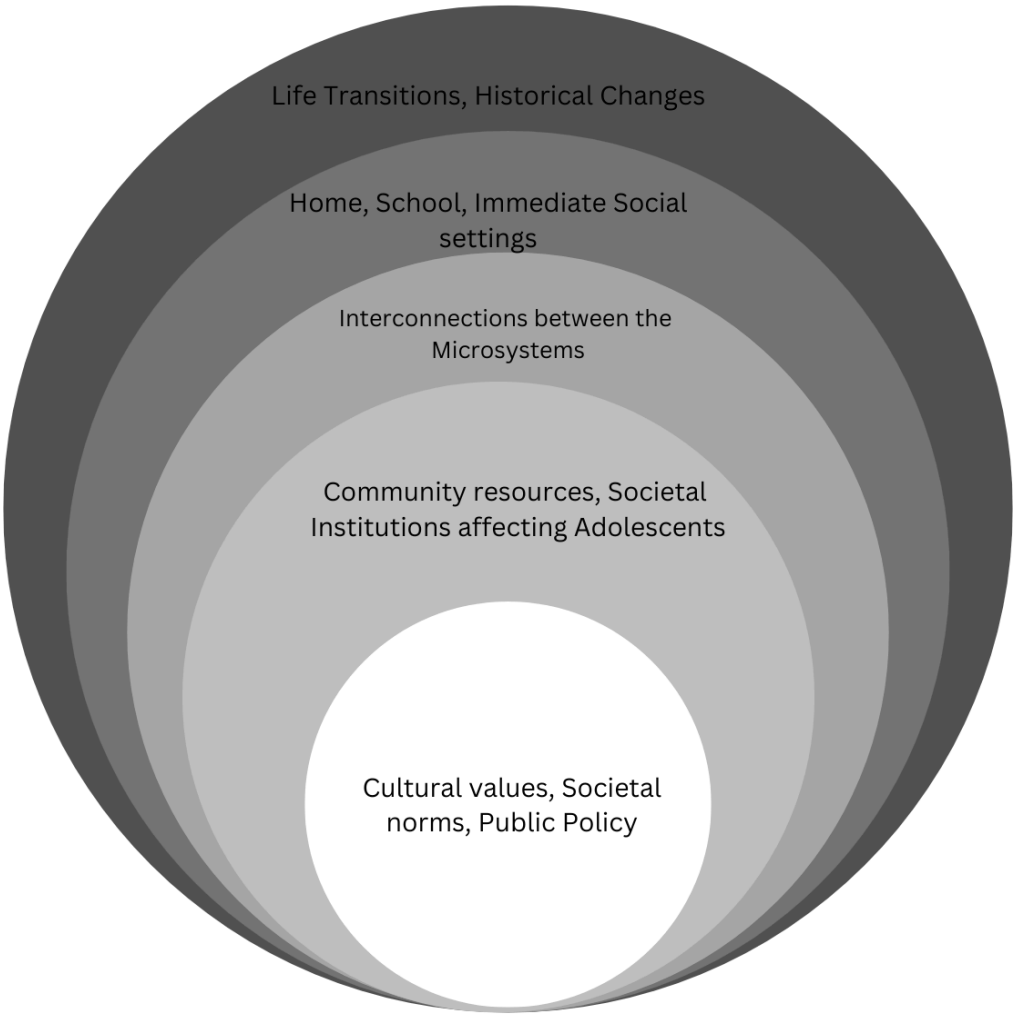
Theoretical Framework

Circadian Rhythm Disruption Theory:
This theory establishes the biological foundation for comprehending the effects of blue light, and how the suppression of melatonin disrupts sleep cycles. Acknowledging this relationship is crucial for addressing challenges related to sleep and anxiety.

Stress and Coping Theory:
The framework proposed by Lazarus and Folkman clarifies adolescents' psychological responses to the impacts of blue light. Emphasizes how beliefs regarding these effects contribute to stress levels. Implementing effective coping strategies, such as utilizing blue-light filters or reducing screen time, presents practical avenues for enhancing mental health.

Integration:
This study emphasizes the complex interplay between sleep, mental health, and technology, integrating biological and psychological frameworks. These insights are vital for formulating interventions that comprehensively address the physiological and emotional dimensions of blue light exposure, thereby ensuring holistic support for adolescent well-being.

Conceptual Framework

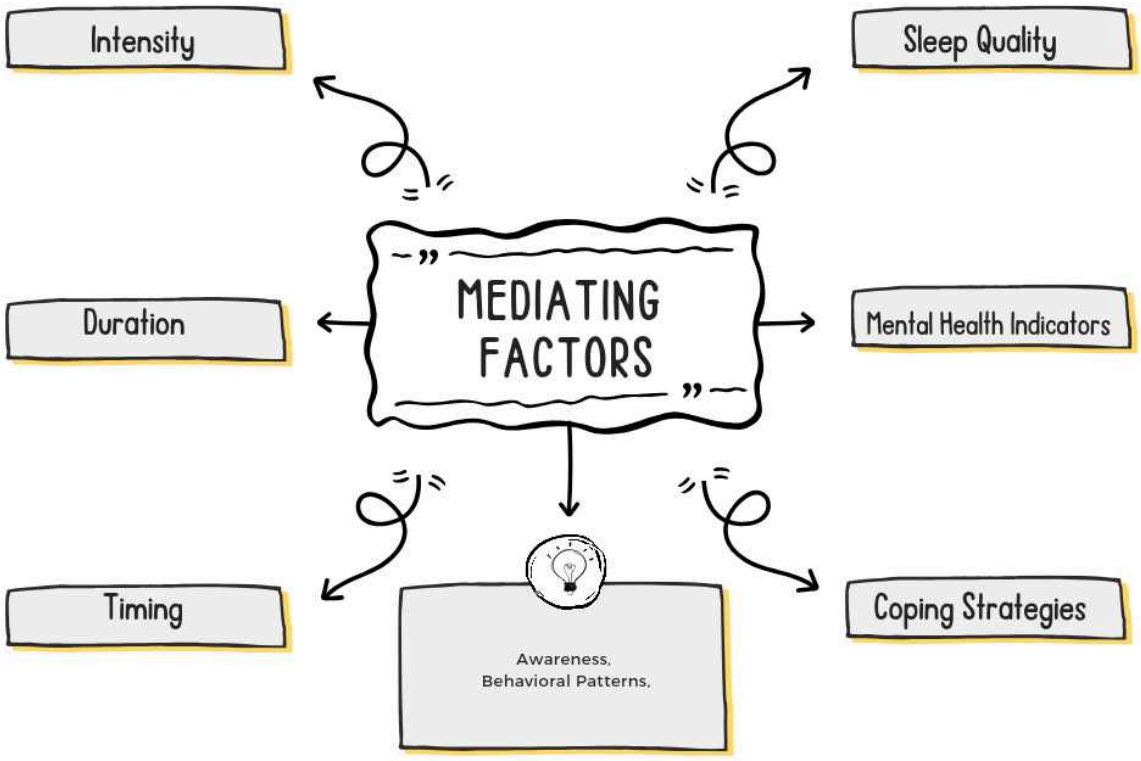


1. **Microsystem:**
   Direct interactions within adolescents' environments play a critical role in shaping their health and well-being. For example, the frequent use of electronic devices at home and in educational settings exposes adolescents to blue light, while peer behaviors significantly influence their device usage during late-night hours.
2. **Mesosystem:**
   Effective communication between parents and teachers exemplifies how external guidance affects technology usage patterns. For instance, discussions surrounding the necessity of balancing technology use for educational purposes with the importance of

maintaining sleep hygiene underscore the significance of collaborative adult supervision in mitigating adverse consequences.

3. **Exosystem:**
   Broader social contexts exert indirect influences on adolescents' habits and perceptions. Initiatives within the community focused on technology education and mental health awareness campaigns play a substantial role in shaping societal norms and behaviors.
4. **Macrosystem:**
   Dominant cultural and societal values of technology—such as its integration into daily life and policies advocating for digital well-being—considerably shape adolescents' attitudes toward screen time and sleep hygiene. Emphasizing the correlation between cultural beliefs and individual behaviors highlights the necessity for evidence-based societal transformations.
5. **Chronosystem:**
   Temporal factors, including the lasting impact of the COVID-19 pandemic on increased screen usage, illustrate the evolving challenges associated with technology integration. Examining the intersection of historical changes and life-stage transitions (for example, during adolescence) with exposure to blue light offers a dynamic perspective for devising long-term solutions.

Conceptual Paradigm



Significance of the Study

Understanding how teenagers perceive the impact of exposure to blue light is a critical step toward understanding larger concerns about their health and well-being. This study could provide information for developing interventions in the form of educational programs, better sleep hygiene practices, and the adoption of blue-light-filtering technologies. The findings could also serve as a starting point in mitigating mental health risks, enhancing academic performance, and promoting emotional stability during the critical developmental phase.

Scope and Delimitations of the Study

The study focuses on five STEM students aged 15 to 16 from a single educational institution narrowing the scope for an in-depth exploration of individual experiences. The limitation begs the restriction to generalize for broader populations. The study excludes younger adolescents, adults, and non-STEM students, as the target demographic was chosen based on the vulnerability or prone capabilities STEM students have from blue light exposure due to academic and developmental Factors. This study was also only conducted within a month and may have a bias due to the personal experiences of the participants, implying not all STEM students aged 15 to 16 will agree to these responses and have their responses and approaches concerning Blue Light and its effects.

Definition of Terms

1. *Blue Light:* A type of visible light with a wavelength between 380 to 500 nanometers(nm) Containing the shortest wavelength and highest energy of all visible lights. This affects humans' CIrcadian and Melatonin levels and has twice the amount of damage done by green light to the human retina. Some sources may be sunlight or artificial sources such as fluorescent lights, LED TVs, computer monitors, smartphones, and tablet screens.
2. *Circadian Rhythm:* HUman's natural 24-hour cycle is called at times the "Body's clock" as it controls sleep, body temperature, hormones, and appetite, One disorder one can attain this is the sleep-wake cycle disorder when the body's internal clock is out of sync with the environment.
3. *STEM students:* Students pursuing in the field of Science, Technology, Engineering, and Mathematics.
4. *Sleep Quality:* A measurement of how well a person sleeps determined by how long, how they sleep, and how they feel after waking up.

CHAPTER II

LITERATURE REVIEW

The Sleep Quality of each generation becomes more apparent with the evolution of technology to have been greatly affected disrupting their Circadian rhythm or sleep cycles, disrupting their physical and mental health. For many students modern day it is common to sleep gaining less than 7 hours the minimum required for young adolescents as their bodies grow. With the new century, there has been enlightenment towards the mental health of society more people being concerned for their and their peers being more aware day by day. These are 15 studies or articles that will help provide insights, or deeper pieces of evidence to prove how important it is to be aware of the effects of Blue Light Exposure on sleep quality and mental health in adolescents, and how this may improve future generations and current ones.

An Article by Marcia Ines Silvani and the Faculty of Medicine in Switzerland, and Robert Werder and Claudio Perret both Sports medicine institution professors in Switzerland about The Influence of Blue light on sleep, performance, and well-being in young adults as a systematic review discusses how the increasing use of blue light-emitting electronic devices effects of blue light exposure on sleep and mental health are scrutinized. This systematic review outlines how blue light impacts physiological processes, and how the melatonin secretion process for the regulation of sleep and mental health in general is deeply crucial, especially for adolescents who are in their developmental stage with increased device use and are highly susceptible to these effects. The main findings are that evening exposure to blue light delays sleep onset and reduces overall sleep duration, resulting in increased fatigue and impaired cognitive performance the preceding day. The review further discusses how disordered sleep can worsen mental health problems such as anxiety and depression in adolescents, creating a vicious cycle of poor sleep and psychological distress. The body of the research contained increased screen time, especially before bedtime correlating with poor sleep quality. The review cites evidence that shows adolescents who report high levels of screen time are more likely to suffer from sleep disturbances and mental health challenges. In summary, the systematic review argues that the physiological disturbances from blue light not only affect sleep but also contribute to a decline in mental health, making it a significant concern for adolescents who are still developing their coping mechanisms. Poor sleep quality as mentioned, can lead to lower academic performance.

and hinder social relationships, perpetuating feelings of isolation and exacerbating mental health challenges. The connection to my study is the theme in regards to the systematic review comes with an important conclusion since adolescents expose themselves more than ever before to the emission of devices. By acknowledging the negative effects of blue light, the possibility of developing measures like blue light filtering apps or limiting screen time at bedtime can be realized. Investigating this relationship could inform interventions aimed at helping young adults develop better sleep hygiene and mental well-being, equipping them to navigate their increasingly digital world more effectively.

The second Article by Prather and HUrst established the foundation of the complex relationship between sleep and circadian rhythms, This showed how these two aspects work together to influence overall health, especially in adolescents who are often subjected to irregular sleep patterns due to lifestyle factors and digital media use. The article focuses on the mechanisms of how sleep and circadian cycles work, exploring how interruptions to one trigger the disturbances of the other. In terms of breadth, authors would sum up some existing literature about what was known by physiological and psychological effects attributed to the alignment or misalignment of sleep and circadian rhythms, then address related factors that impact the overall quality of adolescent sleep such as light exposure-environmental factors that, in their effects, make a real difference. To support this hypothesis, documented evidence from various studies proves the negative impact of low-quality sleep on mental health, especially in adolescents. For instance, they mention findings that associate more screen time, especially before bed, with reduced sleep duration and quality, thus heightening the risk of anxiety and depression. The authors discuss how the blue light emitted from screens disrupts the production of melatonin, a hormone that regulates sleep-wake cycles. Such disruption can cause delay in falling asleep and reduced sleep duration, which can worsen mood disorders common among adolescents. Additionally, the authors discuss potential interventions, such as reducing screen time and using blue light filters, to minimize the effects. In conclusion, findings from the article underscore a relationship between sleep quality, blue light exposure, and mental health, which underlines the need for raising awareness and developing intervention strategies that would improve sleep hygiene among adolescents.

The third article by Wahl, Engelhardt, and three other professors goes into the discussion of the circadian rhythm the inner clock how blue sets the human rhythm. Exposing the body to blue light, which is emitted by all modern light-emitting devices, has become an increasingly important concern because it influences circadian rhythms, mainly in adolescents. This rising use of smartphones, tablets, and computers during the evening hours can disturb sleep and mental health due to the suppression of melatonin levels, delay in falling asleep, and increase in

Alertness at night. These factors are associated with adverse effects on emotional regulation and cognitive performance in this vulnerable group of adolescents, who are already vulnerable due to changes in their circadian rhythms and increased device use. The use of LE devices increases alertness and suppresses melatonin, which further leads to low-quality sleep and high mental health risks. Sleep disturbance in adolescents may contribute to worsening stress, anxiety, and depression. Targeted interventions are therefore required. The findings are a direct reflection of how exposure to blue light directly relates to disrupted sleep patterns, which then cascade into broader mental health challenges. Adolescence, being the period of heightened neurobiological and social change amplifies these risks, which makes preventive strategies like regulating screen use and promoting healthier habits all the more important.

Andre Comiran Tonon and 18 other researchers in Brazil, Washington, and the United Kingdom study bring out the significant association between sleep disturbances, disruptions in circadian rhythm, and the prevalence of major depressive disorder among adolescents. It calls attention to the need to understand how such factors as nocturnal light exposure, particularly blue light, contribute to these mental health challenges among this age group. Introduce the need for there to be at least risk factors that could be used to inform a preventive strategy, because most adolescents have limited access to effective treatments. Artificial light at night, ALAN exposure is associated with great effects on the sleep of adolescents, as well as their mental and cognitive performance. Higher ALAN exposures are related to late bedtime and higher incidence of mood disorders because suppressing melatonin affects the quality of sleep onset. These problems will be compounded among major depressive disorder individuals since their sleeping difficulties include insomnia. Poor sleep hygiene is already a common problem among teenagers, and irregular sleep patterns and nighttime light exposure only make it worse. Blue-enriched light, emitted from devices, delays circadian alignment and reduces restorative sleep. This results in decreased sleep efficiency, heightened risks of depression, anxiety, and stress, and negative impacts on academic and emotional well-being. Furthermore, although blue light can promote alertness and reduce sleepiness during the day, it can impair accuracy and cognitive control during tasks, indicating a trade-off between wakefulness and accuracy. The complex relationship between ALAN, sleep disturbances, and mental health points to the need for targeted interventions that could promote better sleep hygiene and minimize these effects on adolescents' health and performance. The effects of blue light are mediated by its capacity to disrupt the natural sleep-wake cycle. Adolescents, who are already predisposed to delayed circadian phases, face compounded risks from blue light, leading to shorter sleep durations and heightened emotional instability. These disruptions not only undermine mental health but also exacerbate the challenges of academic and social demands during adolescence. The complex connection between blue light exposure and the degradation of sleep quality and consequently mental health

Highlights the growing need for effective interventions as well as educational programs educating adolescents about the risks of nocturnal light exposure particularly in today's technological settings.

The fifth Article about the impacts of blue light exposure from electronic devices on circadian rhythm and sleep disruption in adolescent and young adult students by Mudassir Alam and three other biologists in India begins with the role of blue light in disrupting circadian rhythms and negatively impacting sleep quality. It focuses on how adolescents are uniquely vulnerable to these effects due to their developmental sleep needs. Scientific findings show that exposure to blue light, particularly in the evening hours, suppresses the secretion of melatonin, an important hormone involved in sleep regulation. This impact would lead to poor outcomes, not only in the duration and quality of sleep but also in mental health. An important reason to study this deeper. The supporting evidence identified in the article suggests that the core argument posited by the article—blue light being harmful to sleep quality and mental stability is highly plausible. As such, the physiological grounding of melatonin suppression accounts for the anecdotal reason that exposure to blue light consistently, especially in the evening, causes adolescents' circadian rhythm to be dysfunctional. This is alarming in that sleep plays a large role in the cognitive processes and emotional balance that characterize adolescents. Analysis reveals that even if blue light can boost alertness and cognitive performance in the short term, these advantages are overridden by the negative effects of low quality sleep-considering that adolescents usually operate in stressful environments. Moreover, the mental effects, such as increased anxiety and depression symptoms, that accompany sleep deprivation make interventions reducing exposure to blue light, especially during late evenings, critical. In conclusion, the relationship between blue light exposure sleep quality, and mental health is a concern for adolescents. It may be possible to improve sleep quality by mitigating blue light exposure, thereby enhancing the mental health outcomes of young people.

The sixth article about the 24-hour lighting scheme to promote alertness and circadian entertainment in railroad dispatchers on rotating shifts a field study conducted by PhD Sahin and PhD Figuerio, both light researchers in New York, United States of America abstract outlines a study focused on improving the sleep quality and circadian rhythms of shift-working railway personnel who are prone to fatigue as a result of disruptions to their light-dark exposure. The main objective of the research was to assess the effectiveness of a lighting intervention designed to realign circadian rhythms, improve sleep quality, and reduce sleepiness, which ultimately impacts performance in safety-critical roles. The study was specifically focused on the effects of various lighting interventions (a combination of red, blue, and white light) to correct circadian misalignment resulting from shift work. The study aimed to achieve four goals: (1) improve the

Alignment of the light-dark cycle with natural biological rhythms (2) advance the onset timing of melatonin production in low-light conditions(3) enhance both objective and subjective sleep quality(4) decrease subjective sleepiness, especially during off-duty hours. This successfully introduced greater synchrony between day-shift workers' light exposure and their rest-activity patterns. Although this did not advance the timing of the onset of melatonin, it was associated with lower sleep disturbances and had a positive, albeit nonsignificant, effect on subjective sleep quality. Of critical importance, subjective sleepiness was reduced while off-duty during the day shift. Although none of the desired outcome measurements advancement of the onset of dim light melatonin had been fully met. While the intervention appeared promising as improving sleep quality and off-duty sleepiness, all of the overall effects promoted circadian alignment rather than improving sleep onset timing and quality during duty hours directly.

The seventh Article relates to the effects of evening smartphone use on sleep and declarative memory consolidation in male adolescents and young adults conducted by Christopher Höhn and four other peers providing how the exposure to blue light from digital screens has increased significantly with the widespread use of smartphones and other LED-based devices. Teenagers are more vulnerable as they are in their development stage and often use such devices near bedtime. Such disruption of natural circadian rhythms can have effects on not only sleep quality but also broader aspects of mental health, including emotional regulation and cognitive functions. Supporting evidence from the article showed how melatonin effects are suppressed due to blue light and how adolescents recover faster than young adults. How at bedtime adolescents had normal melatonin levels but adults were still suppressed, and how neither group exhibited significant effects on sleep-dependent memory performance in their research. The analysis based on this is how age-related differences in physiological resilience in adolescents may be overcome by the quicker recovery of melatonin. How adult sensitivity suggests longer recovery times and a potential for greater impacts on deep sleep stages. Finally how memory consolidation is still maintained despite melatonin or sleep-wake cycle changes. These results are consistent with guidelines for sleep hygiene that limit screen time before bed. The data indicates age-specific differences that may be used to create age-specific recommendations for teens and young adults.

The eighth article focuses on different sleep patterns during the COVID-19 pandemic and its association with mood, exercise, and light exposure, written by Mirian David, Gilberto Vieira, and nine other researchers. The study aimed to assess the sleep-wake patterns, mood, perceived stress, and behaviors of physical exercise and exposure to natural light among college students during the COVID-19 pandemic. The study took place between June and August 2020 and addressed how these factors affected the mental health and sleep quality of students. The

The importance of sunlight exposure and physical activity in good sleep hygiene and mental well-being during a global health crisis is pointed out by the findings. Students with higher sunlight exposure and exercise levels showed better quality sleep. Students who suffered from high daytime sleepiness and low sleep quality also had increased levels of anxiety, depression, and perceived stress; therefore, there was a relationship between disturbed sleep and mental health. These aspects were worse in women who had bad quality sleep and more problems in mental health. This study gives a comprehensive overview of how environmental and behavioral factors affect sleep and mental health during a stressful time such as a pandemic. It indicates that not all students have the same sleep disruption but rather that individual habits like sunlight exposure and exercise could buffer the negative impacts of poor sleep. The association between poor sleep and higher levels of anxiety and depression reinforces the importance of addressing sleep quality as part of mental health care, at least during such crises, as the COVID-19 pandemic.

The ninth article by Hohman and Wong on sleep quality and cognitive performances role to whether quantity matters discusses the critical relationship between sleep quality, quantity, and cognitive functioning, especially among adolescents. It stresses that adequate sleep is crucial during this developmental stage due to significant physical and emotional changes that occur. It underscores the importance of both the quality and duration of sleep to achieve optimal cognitive performance and overall well-being. Results show that poor sleep quality leads to undesirable outcomes concerning attention, memory, and emotional stability. Some studies documented positive correlations with improved sleep quality and higher cognitive scores, especially among adolescents that enable them to achieve more in school. The article provides interventions for better sleep habits, such as a need for educational systems and parents to encourage sleep hygiene, which will improve the cognitive performance of adolescents as well as their health in general. The article significantly throws light on the adverse outcomes of sleep deprivation, framing those outcomes within the larger scope of adolescent cognitive and emotional development. It effectively merges quantitative data with qualitative assessment, focusing on the need for more targeted interventions toward improving the quality of sleep in adolescents.

The tenth article by Gibbons and Rieger is a literature review of the impact of circadian rhythms on sleep patterns. It explains how Circadian rhythms are intrinsic 24-hour cycles that play a very important role in sleep and many other physiological processes. Provides extensive insight into how disruptions of these rhythms impact sleep quality and health in general. It is thus essential to understand the association between circadian rhythms and sleep, particularly when factoring in extrinsic factors such as blue light exposure, which has become rampant in modern life, especially among adolescents and young adults. Research shows that exposure to blue light at night seriously compromises sleep quality by delaying the secretion of melatonin and altering it.

Circadian rhythms. It is also established that adolescents are more susceptible to such impacts because their natural circadian rhythms already experience shifts in the process of development. Systematic reviews indicate that a greater percentage of adolescents suffer from sleep disturbances caused by exposure to blue light than adults. Research shows that blue light does not only reduce the amount of sleep but also affects the quality of sleep because it increases sleep latency and reduces sleep efficiency. These same disruptions in this group of individuals, given blue light exposure often associated with delayed sleep timing and lowered melatonin levels, might exacerbate mental health disorders in adolescents, who, already, face high-stress academic pressures and the dynamics of socialization. The study emphasizes that efficient interventions should target light exposure specifically among populations that are also at risk for both disorders of sleep and related psychological issues. Stress management, education in sleep hygiene, and reduced evening exposure to blue light could create significant benefits both in sleep quality and in mental health outcomes in adolescents.

The eleventh article published about the sleep characteristics across the lifespan in 1.1 million people from the Netherlands, United Kingdom, and United States is a systematic review and meta-analysis written by 83 authors that explains how blue light emitted by electronics has recently received attention concerning its potential adverse effects on our health. It possesses a wavelength that affects the circadian rhythm, thereby affecting sleep patterns and, consequently, general well-being. This systematic review seeks to summarize the available evidence related to the impact of exposure to blue light on sleep quality, performance, and mental health, especially among adolescents who are most exposed to screens. Supporting evidence for the sentence includes various studies that have documented altered sleep patterns and increased reports of insomnia linked to prolific digital screen use among adolescents. For example, a study revealed that increased screen time correlates with delayed bedtimes and insufficient sleep durations among children, raising concerns about the long-term mental health ramifications stemming from disrupted sleep cycles due to blue light. The analysis reveals a dual perspective on blue light exposure: while it can enhance alertness during the day, its evening effects are predominantly negative. This dichotomy suggests the nuances in how blue light interacts with human biology, particularly in a demographic that increasingly relies on technology for education and leisure. The findings imply a need for strategies that balance the benefits of daytime blue light exposure with the necessity of safeguarding sleep hygiene in adolescents. This critical discourse about how widespread electronic use affects not only sleep duration but also mental states, such as anxiety and depression, which are rife in this age group, is underlined by the reviewed literature. An urgent need to address these issues through educational programs and parental guidance on technology use and its implications for sleep and health among adolescents is accentuated.

The twelfth article published by Harvard Health 2020 topic is on the effects of blue light on sleep quality solution being blue light glasses and whether they can improve one's sleep. The possible impacts of blue light on sleep and human health in general make the topic quite important. Light from electronic devices has been known to disrupt sleep cycles and cause interference with one's circadian rhythm among teenagers. Knowing how the effects of blue light will influence the quality of their sleep and, consequently, their mental well-being has become imperative in this growing digital age. The accumulated research demonstrates that though exposure to blue light increases alertness during the day, night exposure to it disrupts sleep. Teenagers being the most vulnerable section to this are at higher risks of being affected cumulatively due to the basic need for them to gain enough sleep to adequately facilitate cognitive and emotional growth in their brains. This interference with sleep due to the growing use of digital equipment compromises their academic performance as well as their mental well-being.

The thirteenth article from Harvard Health once more in 2021 is about the dark side of blue light. Exposure to blue light, especially from electronic devices, has gained much attention because of its significant effect on sleep quality and general health. Harvard Health Publishing points out how artificial lighting, particularly blue light, interferes with the biological clock of the body and can pose risks to health, such as diabetes and heart disease. The increased reliance on artificial light changes our natural circadian rhythm, thereby making sleep-related problems and mental health issues worse, especially in teenagers. Blue light exposure has been demonstrated to affect sleep quality by suppressing melatonin production, which in turn delays the onset of sleep and changes the circadian rhythms. This effect has been associated with an increased risk of mental illness, such as anxiety and depression. However, it has also been found that blue light enhances cognitive performance, including mood, learning, and decision-making, during the day. Systematic reviews indicate a balance is required, especially among vulnerable groups such as adolescents, as blue light is associated with impairing sleep but also enhancing alertness. Analysis of the evidence yields a complex relationship between light exposure to blue and adolescent health. While daytime exposures to blue light increase alertness and cognitive performance, blue light at night is particularly detrimental to sleep quality; it even worsens existing mental health conditions. Among adolescents, circadian rhythms of wakefulness during the day and at night naturally predispose people to be awake at certain times, leading to sleep deficiency, which significantly impacts student performance and emotional stability. Reduce exposure at night to maximize sleep and mental health with appropriate exposure during the day shall enhance cognitive abilities and performance academically. Thus the effective strategy should manage blue light exposure to maximize the beneficial effects of blue light at any time of the day by minimizing the adverse effects through exposure at night. A nuanced approach is necessary to promote healthy sleep patterns and support the mental well-being of adolescents.

The fourteenth article Harvard Health 2019 on whether blue light from electronic devices increases one's risk of macular degeneration or blindness explains how blue light emitted by various electronic devices has generated issues about its possible contribution to eye health problems, more specifically macular degeneration, and blindness. Specialists often have questions about whether exposure might increase the risk for this kind of disease, primarily among smartphone, tablet, or computer users. While blue light does possess certain characteristics that could impact health, this is not directly relevant to macular diseases. Emphasizes that the emission of blue light through digital devices does not pose an imperative threat to the retina as well as an elevated predisposition to macular degeneration. Blue light is emitted more vigorously in natural sources, specifically sunshine, than in everyday gadgets, it is indeed proven that long exposure to display screens in the late afternoon affects sleep as there is a hindrance in the release of the hormone melatonin is responsible for the sleep-wake cycles.

The fifteenth article about how blue affects sleep is written by Mikulska and Skoczylas both researchers in Poland have their article explain the circadian rhythm, which controls sleep patterns, is significantly affected by light exposure, especially blue light. Although blue light is useful for alertness and cognitive performance during the day, exposure to it at night inhibits the secretion of melatonin, causing poor sleep quality. Adolescents are particularly vulnerable to circadian disruptions, and they often have longer exposure to blue light through screens in the evening, which can affect their sleep and psychological well-being. This suggests the need for controlling blue light exposure, particularly at night, for adolescents to improve sleep health. Supporting evidence explains how Blue light entrains the body clock by acting on the SCN, however, night exposure desynchronizes the rhythm. How long prolonged evening exposure to LED screens delays the onset of sleep and impairs the quality of sleep. It also mentions there is inconsistent evidence where some studies indicate decreased sleep duration and efficiency among adolescents who are exposed to blue light before bedtime. Increased latency or wait to sleep and decreased deep sleep(N3 stage) have also been reported in some studies. The psychological effects of the study mention how chronic sleep deprivation due to circadian rhythms may aggravate anxiety and depression in adolescents. The analysis is how the physiological susceptibility of adolescents to blue light is based on their more rapid melatonin production and, therefore, a rather sensitive sleep-wake cycle. The widespread use of LED devices makes the problem acute, as youth frequently employ such devices just before bedtime. These findings outline the need for interventions such as blue-light filtering and behavioral modifications to protect adolescents from the adversity effects of nighttime blue-light exposure on sleep and mental health.

In summary, the articles collectively examine the effects of blue light exposure on sleep quality and mental health in adolescents, emphasizing the biological and behavioral disturbances that occur due to the overuse of electronic devices during the evening. Some key findings include the suppression of melatonin production, delayed sleep onset, and a general decrease in overall sleep quality, resulting in increased sleep latency and decreased restorative sleep. This disruption exacerbates mental health issues, such as anxiety and depression, that are already common in the adolescent developmental stages. During adolescence, natural circadian rhythms tend to be phase-delayed, making adolescents vulnerable to the adverse effects brought about by late-night screen exposure. Moreover, other literature indicates potential interventions through blue-light filtering technology or setting curfews for devices to minimize these negative impacts.

Studies emphasize the two-way relation between poor quality of sleep and mental health, revealing that poor quality of sleep not only degrades cognitive and emotional functions but also sets a cycle of chronic sleep deprivation with psychological stress. The above findings urge for targeted intervention in exposure to blue light, as this may further enhance sleep hygiene and adolescent mental health in an increasingly digitalized world.

CHAPTER III

RESEARCH METHODOLOGY

Seeks to explore the influence of blue light exposure on sleep quality and mental health in adolescents and understand their perceptions, behaviors, and coping strategies. Recapping to know how to respond and help the student's research seeks to answer the following:

1. How do STEM students individuals aged 15 to 16 perceive the impact of blue light exposure on their sleep quality and emotional well-being?

This is the main question aimed to get an idea of the subjective interpretations and personal experiences related to the issue of blue light exposure and its impact on sleep quality and emotional well-being during adolescence. Striving to identify whether adolescents are aware that their exposure to blue light influences their circadian rhythm, sleep quality and duration, and other secondary outcomes on mental health by impacting mood, stress levels, and emotional regulations listed within this study.

1.1. What are the participants' personal experiences regarding the effects of screen time before bed?

1.2. How do participants describe changes in their emotional state related to their screen time habits?

1.3. What coping mechanisms do participants use to manage blue light exposure and its impacts on sleep?

2. (Awareness) How do STEM students aged 15 to 16 perceive the existence of blue light and its impact? What do they believe or think about it and its role in affecting sleep and mental health?
3. (Behavioral Patterns) How do STEM students aged 15 to 16 use blue light-emitting devices such as smartphones, tablets, or computers in the evening? What is their habit or routine that would increase the exposure?
4. (Coping Strategies) What coping strategies do STEM students aged 15 to 16 use to reduce their exposure to blue light, including blue light filters, limiting screen time at night, or participating in other sleep hygiene behaviors?

This investigation will gain an understanding of how youth conceptualize and cope with the advancements in technology use, sleep health, and emotional well-being, which in turn could help highlight the need for educational or technological-bad interventions to address these.

Design of the Study

This qualitative study employs semi-structured interviews and open-ended surveys to explore adolescents' experiences with blue light exposure. "Research is formalized curiosity. It is poking and prying with a purpose," by Zora Neale Hurston(1891-1960). Qualitative research is a method focused on understanding human behavior, experiences, and social phenomena through non-numerical data, involving more collecting and data such as interviews, focus groups, observations, and textual materials to explore the "why" and "how" of a topic.

Population and Sampling

The study involves five STEM participants' ages ranging from 15 to 16 as it is typically the critical development stage from middle to late adolescent transition. At this age group, there is more development in advanced reasoning and reflective capabilities having experienced significant cognitive, emotional, and social changes making this age group insightful for studies on identity formation, decision-making, or peer influence. This is also an age for balance wherein they are young enough to recall early adolescent experiences yet old enough to anticipate young adult challenges especially as the 5 participants are all STEM students for this research topic from the grade 11 year 2024 to 2025.

Research Instrument

The researcher will use a semi-structured interview guide, an open-ended survey questionnaire, and validated instruments like the Pittsburgh Sleep Quality Index(PSQI) and the General Anxiety Disorder-7(GAD-7). The questionnaire will contain two questions per sub-topic (Awareness, Behavioral Patterns, Perceived Effects, and Coping Strategies) to understand and explore the knowledge and insights of the current grade 11 students aged 15 to 16. This portion is to gather more in-depth knowledge about Blue light, whereas the PSQI is to evaluate how much sleep the students had during the former month of November in total, asking questions of how long they stay awake in bed to how long they get the total amount of hours asleep. The purpose is to compare their awareness and how they implement their knowledge. This follows the GAD-7 where the results will determine the effects of sleep deprivation, and answer any other factors involving their emotional distress apart from using their devices in the late evening.

Validation of Instrument

Per identifying the perceived effects and awareness of Blue light effects on STEM students aged 15 to 16, the researcher can find two standard survey tests, the Pittsburgh Sleep Quality Index(PSQI) and the General Anxiety Disorder-7(GAD-7), both having been used many times in the field of psychology and for any civilian as it is published publicly with the scoring graph attached.

Method and Data Collection

The researcher will be using three instruments in total, the semi-structured interview where the respondents answered face-to-face interviews, and on-the-phone interviews, whereas the two surveys were followed the same each session in total lasting approximately 14 to 26 minutes. The questionnaires would then be scored by the researcher and the data of each participant would be compared and analyzed to evaluate the understanding of grade 11 STEM students on blue light effects.

Plan of Data Analysis

To provide nuanced insights from qualitative data, the researcher chose thematic analysis to be chosen. The thematic analysis offers to have the researcher's familiarity with the data as you transcribe from its audio form, take notes on initial impressions, key points, and potential patterns, and highlight significant phrases, sentences, or ideas beneficial to the study. The coding framework was used to identify initial themes and saturation was achieved when no new themes were to emerge ensuring the validity and reliability of findings.

CHAPTER IV
RESULTS

1. **Awareness**

   Participants displayed varying levels of awareness regarding blue light and its effects within their responses.

   - Participant 1: Mentioned blue light improves alertness and memory and how it helps cope with depression but does not have harmful effects. For melatonin

levels and Circadian Rhythms, the participant was unaware of the terms or relevance.

- Participant 2: Knew about blue light causing eye strain and affecting sleep schedules but unaware of broader effects. For knowledge of Melatonin Levels or Circadian Rhythms, the participant was completely unfamiliar with both.
- Participant 3: Aware that blue light distracts from sleep but provided no specific mechanisms. Knew of negative influence on Melatonin Levels but lacked depth.
- Participant 4: Limited knowledge of blue light relating it primarily to screen emissions and general effects on focus. Had no prior knowledge of Melatonin levels or Circadian Rhythms until explanation from researcher.
- Participant 5: Described blue light as part of the visible light spectrum and accurately connected it to melatonin suppression and sleep distribution. Had the most understanding of explaining how blue light inhibits melatonin secretion.

2. **Behavioral Patterns**

Device usage was consistent among participants, with many reporting prolonged screen time, particularly at night with a screen time of 2 to 14 hours a whole day. Activities included gaming, social media, academic tasks, and general entertainment having breaks before bed was rare and most participants used devices until the point of sleep.

- Participant 1: Screen time of 6 to 14 hours; activities included Tiktok, Instagram, Facebook, and Gaming.
- Participant 2: Screen time of 7 to 9 hours of nightly device use; primarily for gaming, social media, and academic research.
- Participant 3: Screen time of 2 to 3 hours nightly; primarily for studying and entertainment(games and video watching).
- Participant 4: Screen time of 5 to 12 hours; most time spent studying or gaming.
- Participant 5: Screen time average of 4 hours; gaming prioritized over social media.

3. **Coping Strategies**

Coping mechanisms varied with some participants employing strategies such as blue light filters or engaging in relaxing activities while others struggled to implement change citing academic pressures or habitual behaviors as barriers, or had no specific strategies in place indicating a lack of prioritization of the issue.

- Participant 1: Relieved stress by gaming, chatting with friends, sketching, or playing the electric guitar; uses a blue light filter always. Other strategies include breathing techniques and daydreaming to relax.

- Participant 2: Did not actively reduce screen time, however, uses blue light filters from 6 pm to 6 am. Claimed hearing strategies such as timers or reducing brightness but does not apply them personally.
- Participant 3: Tried reading books but found it ineffective, gave devices to parents to limit access; has blue light filters on always.
- Participant 4: Had not tried reducing screen use, citing the need for academic work, but suggests reducing academic workload and screen time to improve sleep; uses blue light filters continuously.
- Participant 5: Suggested keeping devices away but had not implemented significant changes or use of any blue light filters. No further strategies not having considered blue light exposure a problem.

4. **Sleep Quality**

The common theme among participants with delayed sleep onset, difficulty staying asleep, and poor rest was reported. Many had acknowledged that extended blue light exposure contributed to these problems, though few believed their sleep issues were unrelated to device use.

- Participant 1: Experienced delayed sleep due to overthinking and procrastination, Reported being unresponsive and slow due to lack of sleep. PSQI score of 43%
- Participant 2: Acknowledges disrupted sleep's impact on stress and susceptibility to illness. PSQI score of 47%
- Participant 3: Reported shorter temper and frequent frustration from sleep difficulties. PSQI score of 38%
- Participant 4: Noted discomfort and lack of energy following poor sleep. PSQI score of 38%
- Participant 5: Claimed insomnia was independent of device use but admitted frustration and mood changes from disrupted sleep. PSQI score of 29%

5. **Anxiety Indicators**

Participants often link disrupted sleep to heightened anxiety, frustration, or irritability with some mentioning overthinking or stress during nighttime device usage, and emotional responses like restlessness and low energy were frequently associated with poor sleep and extended screen time.

- Participant 1: GAD-7 score of 17 out of 21
- Participant 2: GAD-7 score of 17 out of 21
- Participant 3: GAD-7 score of 13 out of 21
- Participant 4: GAD-7 score of 18 out of 21
- Participant 5: GAD-7 score of 3 out of 21

CHAPTER V

DISCUSSION

The findings underline the urgent need for public health interventions aimed at reducing blue light exposure, particularly in the hours leading up to sleep. Educational initiatives should focus on improving awareness of blue light's effects and promoting evidence-based strategies, such as limiting evening screen time and using blue light filters.

Conclusion

This qualitative research will explore the relationship between blue light effects on STEM student aged 15 to 16 on their sleep and mental health. The results of this study will contribute to the existing literature relating to the research or studies of psychological health issues. This study will help the students to understand that blue light is a factor and to lessen their screen time in correlation to having enough sleep to prevent lack of bodily functions in the day, negative moods, and decrease the chance of anxiety. Teachers and parents will also be able to benefit from this study because they can understand their students or children more on a deeper level of their concerns at an academic level and human perspective, to then find ways to reduce struggles for future generations on their mental well-being. The school itself will also be concerned about seeing the result of this study because they would be more aware of their student's mental health and adjust the curriculum or find other options. Moreover, the results of this study would help in providing a wider understanding of the connections between devices to blue light effects on sleep to how our brains function specifically those ages 15 to 16.

Recommendations

Further research done by the researcher or others should explore the interplay of individual factors such as coping styles and resilience, in moderating the impact of blue light on sleep and mental health. The study number of participants should stretch out to other schools to have more diverse populations and STEM in comparison to non-STEM students, as well as turning into a comparative study on sleep and blue light effects, and address further barriers to behavior change such as academic demands and ingrained habits.

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Transcription

Respondent 1

Interviewer: Today is November 28, 2024. This is Practical Research 1, the Influence of Blue Light Exposure on Sleep Quality and Mental Health in Adolescents. First interviewee,----------. So, the first question here is awareness. You have around as long as you feel to answer the question right now, as long as we can finish it within today okay? so, what do you know about blue light and its effects on sleep and mental health?
Interviewee: Hold on, hold on, hold on.
Interviewer: Okay I'll repeat, what do you know about…
Interviewee: -girl-
Interviewer: about the light and it effects on sleep and mental health?
Interviewee: All I know is that it improves your alertness. It can also boost your memories and it can help with you coping with depression. Because sometimes you do feel…you tend to feel lonely and the only way you cope is using your device. You can message with online friends through games or other apps, yes.
Interviewer: Okay, thank you very much next question is well um what have you ever heard about blue light influences, melatonin levels or circadian rhythms? So can just say you don't know if you don't know this one .
Interviewee: Yeah I don't know.
Interviewer: Okay thank you very much, it's okay, next one is for behavioral patterns how often do you use electronic devices at night and for how long?
Interviewee: ohhh.
Interviewer: you may check your phone if you if you want.
Interviewee: I forgot when but my usual screen time is like 14 hours *Laughs*
Interviewer: Oh okay.
Interviewee: Yes 14 hours, but I can go like as early as six hours or seven.
Interviewer: K that's early, Okay. I'll just pause this temporarily… And we are back to resume in 3, 2, 1. So, once more, how often do you use your electronic devices at night and for how long?
Interviewee: Normally, I'd go for like 6 hours or up to 14 hours if it's weekends or summer break but the earliest I can go is from four hours or one hour.
Interviewer: okay um what type of activities such as social media gaming and studying do you typically engage in on your devices?
Interviewee: do I have to be specific with this?
Interviewer: Yes please.
Interviewee: So, TikTok, yes. I go scrolling through TikTok. I check latest posts on Instagram and Facebook. I often open Messenger to chat with my friends and family and play games most of the time yes.

Interviewer: Okay Thank you. Do you take any breaks from screen use before going to bed?
Interviewee: Like what do you mean?
Interviewer: Like do you ever prior to sleeping cut your phone off and then do other things such as brushing your teeth, washing up or just not cutting it on before going to bed?
Interviewee: Ah yes, I usually wash my face and brush my teeth before going to sleep, yes.
Interviewer: Okay, then we will then be moving on to perceived effects. Do you feel that using devices at night affects your ability to fall asleep or stay asleep?
Interviewee: Yes. It's often caused from procrastinating, especially in research so I just mostly use it- use my phone every night like overthink until I eventually fall asleep at around four o'clock or five yes
Interviewer: okay have you noticed any changes in your mood stress levels or energy after a night of disrupted sleep?
Interviewee: oh yes um I think cramming and maybe sometimes I go unresponsive. When someone asks me a question, I'm slow to respond or react to it.
Interviewer: That's okay. That's okay. What emotions do you associate with sleep difficulties? Do you have frustration, anxiety?
Interviewee: I feel nothing. Like you're in a deep thought. That's it.
Interviewer: Okay. Okay.
Interviewee: If you get it bruh.
Interviewer: Thank you. Thank you. What are your coping strategies? Have you tried any methods to reduce your screen use at night? If yes, which ones?
Interviewee: Okay, I talk with friends, play with friends in-game. Yes, I have four friends in-game and we usually play for hours.
Interviewer: How does that reduce your screen time?
Interviewee: Playing with them is like a stress reliever because I get to share my problems and how my day was. And they would do the same thing.
Interviewer: Okay, I'll repeat the question once more, thank you. Have you tried any methods to 'reduce' your screen use at night?
Interviewee: Oh!
Interviewer: Yes, yes, sorry, to reduce your screen use...
Interviewee: My bad, my bad, my bad. Okay, so take note that's the effect of lack of sleep.
Interviewer laughs
Interviewee: Okay. Uh, what again…?
Interviewer: Oh, okay. Have you tried any methods to reduce your screen use at night?
Interviewee: Oh, yes. I sketch.
Interviewer: Okay. You draw?
Interviewee: Yes, and play my guitar.

Interviewer: Very talented. Do you use blue light filters or apps to manage screen exposure in the evening?
Interviewee: Eye Comfort?
Interviewer: Yes, that's it.
Interviewee: Yes, yes, yes, I do.
Interviewer: And then for your final one, it is, what other strategies have you heard about or tried to improve your sleep or mental health in relation to screen use?
Interviewee: The serious way or the unserious way?
Interviewer: Um, serious way?
Interviewee: Okay. Breathing technique.
Interviewer: Okay.
Interviewee: Yes.
Interviewer: Okay, Demon slayer I see you
Interviewee: Hmm. Yes. Or maybe, oh no, maladaptive day breathing. *laughs*
Interviewer: Oh, okay.
Interviewee: Yes, until I fall asleep. Just create stories before going back to sleep.
Interviewer: Thank you very much. We will now be moving on to the written test.

Respondent 2
Interviewer: This is interview number two. Today is November 30th. Where the interviewee is --------- and yes. So, the interview of Blue Light Exposure on Sleep Quality and Mental Health in Adolescents with the witness of ----------. We will now begin the questioning starting with awareness. What do you know about blue light and its effects on sleep and mental health?
Interviewee: As far as I know, blue light is like, you know it kind of makes your eyes hurt, because especially like if you're using gadgets for a long periods of time like say like more than five hours in my experience and although blue light is like kind of a threat to your eyes and then like if your eyes are you know sore can affect like your sleep schedule because they will hurt when you it's time to go to bed.
Interviewer: Okay thank you, have you ever heard about blue light influences on melatonin levels or circadian rhythms?
Interviewee: um for me personally I have not heard of that? no I haven't.
Interviewer: Melatonin levels are like your sleep levels and then circadian rhythms are for your emotions.
Interviewee: What again? The last part?
Interviewee: Emotions?
Witness: That's also like your sleep cycle.
Interviewee: Okay.

Interviewer: Still nothing?
Interviewee: No.
Interviewer: Okay.
Witness: Give him time to process.
Interviewee: What?
Interviewer: Okay, next question?
Interviewee: Yeah.
Interviewer: Okay.
Interviewee: You asked me if I know anything, I said no.
Interviewer: Shhhhh We don't need that part. I need to cut that out. (interviewee mumbling) So behavioral patterns, how often do you use electronic devices at night and for how long?
Interviewee: I use like yeah I use my phone every night along with my computer. I always actually use it like right before I sleep for the past I'd say six months yeah I've been using it like right before So basically, every night I'm always on my device.
Interviewer: For how long?
Interviewee: How long? Until I sleep. So I sleep around usually between 12 to 2. So assuming I get home at like 5.30, that's going to be like 6, 7, 8, 9, 10, 11, 12, 3, 2, 1. (10 hours) Like 7 to 9 hours.
Interviewer: Okay. What type of activities such as social media, gaming, or studying do you typically engage in on these devices?
Interviewee: Recently, I've been engaging a lot on Google Docs and Google Drive to view my PDFs regarding the formats for my research. And for gaming, I guess I do play other AAA games such as maybe Valorant and other FPS games. And for social media, Instagram, Facebook, YouTube, I use some of them.
Interviewer: Okay, so all of the above. Do you take any breaks from screen use before going to bed?
Interviewee: No, I don't. I don't. Yeah, I don't.
Interviewer: Okay. Next is for the perceived effects. Do you feel that using devices at night affects your ability to fall asleep or stay asleep?
Interviewee: To a certain degree, yes, I would agree. But there are times where I'm really tired, so I would just fall asleep even with my phone open.
Interviewer: Okay. Have you noticed any changes in your mood, stress levels, or energy after a night of disrupted sleep?
Interviewee: Yeah, if I did have a lack of sleep the night prior, there would be times where I'm not really myself. So when you have a lack of sleep, you're more prone to stress or a headache, and sometimes you might even get sick because of that.

Interviewer: Okay, thank you what emotions do you associate with sleep difficulties such as frustration anxiety and alike
Interviewee: wait just to clarify so like emotions that like interrupt my sleep?
Interviewer: um more like emotions you associate with sleep difficulties.
Interviewee: so like emotions that give me a hard time to sleep?
Interviewer: no, emotions you associate with sleep difficulty like when you think of the term sleep difficulty what emotions do you normally associate or conclude are similar like with such a term.
Interviewee: I guess frustration because like you know when you wake up from like a night but then like you just barely slept you're gonna be like super tired throughout the day so like doing anything is gonna be like more difficult and then like especially if you have a lot of work that's gonna be like very stressful so overall like just negative emotions in general.
Interviewer: Okay thank you then for your terms of coping strategies have you tried any methods to reduce your screen use at night if yes which one?
Interviewee: No.
Interviewer: oh *witness laughs* okay do you use blue light filters or apps to manage screen exposure in the evening
Interviewee: oh on my phone yes I do have like this thing scheduled like where it turns on a blue light filter like Around like when the sun sets so 6 a.m.
Interviewer: 6am?
Interviewee: No no I- like 6 a.m. It turns off, 6 p.m. It turns on, and then like yeah I recently just like applied that to my computer as well.
Witness: Yea good.
Interviewer: Then what other strategies have you heard about or tried to improve your sleep or mental health in relation to screen use?
Interviewee: When it comes to strategies, although I don't really apply them to myself, I've heard like some people, they set like a timer on their phone or like how much they can use it. So like once the timer expires, they just can't use their gadget. I guess there are like other like, you know, alternative methods, but they're not really that effective.like just putting your brightness on the lowest and maybe avoiding flashing colors especially like when you're in a dark environment but I know like in this case like the best like way to just you know avoid blue light is just to stop using gadgets in general.
Interviewer: okay thank you so this ends the first portion where these questions have aimed to gain qualitative data on how adolescents perceive experience and adjust the impact of blue light on their daily lives, while also providing insights for actionable solutions. We will now be taking a temporary break for this.

Okay, so this is the continuation where we are now going to the GAD-7,the generalized anxiety disorder screener, where the interviewer will be answering from the basis of four different options. Okay, so over the last two weeks, how often have you been bothered by the following problems? Such as feeling nervous, anxious, or on edge. Zero is not at all. One is several days. Two is more than half the days. And three is nearly every day-
Interviewee: -three.
Interviewer: Okay. And then number two, not being able to stop or control worrying.
Interviewee: Three.
Interviewer: Worrying too much about different things.
Interviewee: Three.
Interviewer: Trouble relaxing.
Interviewee: This is like in the past two weeks, right? Three.
Interviewer: Okay. Being so restless that it is hard to sit still?
Interviewee: I guess two.
Interviewer: becoming easily annoyed or irritated?
Witness: Three, three, three.
Interviewee: Hey- One.
Interviewer: Oka-
Witness: Three three three.
Interviewee: Okay two.
Interviewer: Feeling afraid as if something awful might happen?
Interviewee: Mmmm One.
Interviewer: *writing scoring* Okay, If you checked off any problems, how difficult have these problems made it for you to do your work, take care of things at all, home, or get along with other people?
Interviewee: What? What am I going to do there?
Interviewer: Is it not difficult at all, for example, if you have problems with how you do work, take care of things at home, or get along with other people, not difficult at all? Somewhat difficult, very difficult at all somewhat difficult very difficult or extremely difficult?
Interviewee: I…don't get it.
Interviewer: okay so um due to your anxiety possible anxiety um how difficult has dealing with your work or taking things at home or getting along with people socially been for you?
Interviewee: I guess work, research, bad.
Interviewer: Is all of those in general, however, in consideration, is it not difficult, somewhat difficult, very difficult, or extremely difficult, including as well your social life?
Interviewee: I don't know. It's only, I guess, work that I'm having an extremely difficult time, but, like, with others, not really.

Interviewer: So?
Interviewee: Just work. with others not really so just work.
Interviewer: …Somewhat difficult, very difficul…
Interviewee: are you like routing them all into like one general-
Interviewer: yeah.
Interviewee: I guess somewhat difficult.
Interviewer: Okay and when do you think that these symptoms had possibly began?
Interviewee: What?
Interviewer: These symptoms have possibly began, all of things you checked off.
Interviewee: When did- when did um midterms end? When was that?... November...
Interviewer: Isn't that October?
Interviewee: Not October, I wasn't that worried with concept paper it was just when the actual paper was being made okay so around like second week of November.
Interviewer: Okay one second week okay we will now be then getting into the next one this is the Pittsburgh sleep quality index so for instructions the following questions relate to your usual sleep habits during the past month 'only', your answer should indicate the most accurate reply for the majority of days and nights in the past month please answer all questions during the past month when have you usually gone to bed?
Interviewee: the earliest I went was like well the latest I went was like 3:30 so I'd say between 12 and 3:30.
Interviewer: Okay. How long in minutes has it taken you to fall asleep each night?
Interviewee: Between 5 and 20 minutes, I'd say.
Interviewer: Okay. When have you usually gotten up in the morning?
Interviewee: Uhh I usually wake up at 7, I'm supposed to, but recently there are times where I just sleep through my alarms, so I've been awake. So like- there was like a day or two where I woke up at like 9.
Interviewer: how about 8?
Interviewee: 8:45, that was only like two days off like the past month.
Witness: Shouldn't you put majority?
Interviewer: yeah I should but I'm going to get the average because he's taking ten years and honestly speaking um his sleep schedule is way worse than what he's saying when he wakes up how many hours of actual sleep do you get at night? This may be different from the number of hours you spend in bed.
Interviewee: Uhh like an approximate number?
Interviewer: Yeah…approximate…
Interviewee: I'd say between…3 to 6 hours.

Interviewer: Okay. During the past month, how often have you had trouble sleeping because you cannot get to sleep within 30 minutes? Zero is not during the past month, One is less than once a week, two is once or twice a week, three is three or more times a week.
Interviewee: I guess one sometimes yeah cause like- like usually sometimes I would just end up like- doom scrolling while I'm in bed so I don't sleep.
Interviewer: uh wake up in the middle of the night or early morning?
Interviewee: Oh no, that never happens to me when I sleep, I sleep.
Interviewer: have to get up to use the bathroom?
Interviewee: nope, when I wake up in the morning I guess? but that doesn't count I think.
Interviewer: Cannot breathe comfortably?
Interviewee: No.
Interviewer: You don't have that? Cough or snore loudly?
Interviewee: Snore I guess *Laughs*
Interviewer: Less than once a week? Once or twice a week? Three or mor-
Interviewee: Everyday.
Interviewer: Oh okay, feel too cold?
Interviewee: Mmm no.
Interviewer: *long pause* could you like- give me an answer *witness incoherent words* from these? It's just yea-
Interviewee: No. I-i i don't feel cold
Interviewer: Okay?
Interviewee: I guess some weeks I guess, I would use a blanket but usually I just need two pillows and a cold room then I'll fall asleep.
Interviewer: Uh feel too hot? *witness yawns*
Interviewee: feel too hot? yeah sometimes.
Witness: He is the direct opposite of me.
Interviewer: Is that once or twice a week three or more times…
Interviewee: I'd say like once or twice a week.
Interviewer: Have bad dreams
Interviewee: None at all. What do you consider bad?
Interviewer: That's something you…define?
Interviewer: Sometimes I have no dreams like I just don't remember. I just fall asleep zzzpp and then wake up.
Interviewer: Have pain?
Interviewee: Pain?..what- like in sleep?
Interviewer: Yeah.
Interviewee: No.
Interviewer: Then for if any other reasons, please describe including how often you had trouble sleeping because of these reasons.
Witness: (researchh)

Interviewee: What?
Interviewer: So this one is something you can do yourself, which is another reason that you have. Please describe um how often you have such trouble sleeping because of this reason.
Interviewee and Witness: Research.
Interviewee: *long pause*Just research.
Interviewer: Okay. So during the past month, how often have you taken medicine, prescribed or over the counter to help you sleep?
Interviewee: Naur I don't take medicine for sleep.
Interviewer: During the past month, how often had you had trouble staying awake while driving eating meals or engaging in social activity?
Interviewee: Trouble staying awake?
Interviewer: Yes.
Interviewee: Oh yeah sometimes.
Interviewer: Once or twice a week less than once a week?
Interviewee: Once or twice a week especially during classes I remember I just slept through the math quiz give what is asked- I like- the moment I saw it I just took a nap when I woke up it was like five minutes left.*laughs alot* It just went right down *incoherent*
Interviewer: I'll be sure to remove that (didn't understand him in recording**) Um during the past month, how much of a problem has it been for you to keep up enthusiasm to get things done? Interviewee:...Enthusiasm? So like- how happy am I?
Interviewer: Yeah.
Interviewee: In general?
Interviewer: Yeah.
Interviewee: Within the past month?
Interviewer: Yep.
Interviewee: Okay.
Interviewer: Is that less than once a week? once or twice a week…not during the past month?
Interviewee: Wait what? Less than once a week am I happy or something?
Interviewer: Are you well um to keep up enthusiasm…like- how much of a problem has it been for you to keep up enthusiasm to get things done?
Interviewee: Ah I'd say once or twice a week.
Interviewer: Okay and then during the past month how would you rate your sleep quality overall?
Interviewee: Bad.
Interviewer: Okay.
Interviewee: Really bad.
Interviewer: Okay s-

Interviewee: Very bad, actually.
Interviewer: I-Uh- Okay. This concludes the interview. Thank you very much.

Respondent 3

Interviewer: I'm here with ---------- on the topic of the influence of blue light exposure on sleep quality and mental health in adolescents. -------, are you present?... ------?
Interviewee: -Yeah.
Interviewer: Owki, so I will read to you the general objectives and then the questions.
Interviewee: Okay.
Interviewer: So the general objective of my study is to explore the influence of blue light exposure on sleep quality and mental health in adolescents and understand their perceptions, behaviors, and coping strategies. If you have any questions in regards, I may repeat the questions or repeat the guidelines for understanding. And then for the specific objectives, it is to assess adolescents' awareness of blue light exposure and its effects on sleep and mental health, to investigate the behavioral patterns of evening use of blue light emitting devices among adolescents, to analyze the perceived effects of blue light exposure on sleep quality, emotion, well-being, and mental health, and then fourth, to identify the coping strategies adolescents employ to mitigate the effects of blue light exposure on sleep and mental health, and finally, to recommend interventions and educational strategies to reduce the negative impacts of blue light exposure. So for my first question, it is, what do you know about blue light and its effects on sleep and mental health?
Interviewee: I only know some aspects of that blue light which is very bad for your sleep and mental health since I've heard that blue light distracts you focusing on sleep.
Interviewer: Okay thank you um have you ever heard…heard about blue light influences on melatonin levels or circadian rhythms?
Interviewee: Yeah, I've heard that blue light influences negatively on those things.
Interviewer: Okay, and how often do you use your electronic devices at night and for how long?
Interviewee: I use my electronic devices usually every day. And I think I use it like for two to three hours. Then I have to finish things and I use it, I use my phone for me to sleep.
Interviewer: Okay, and what type of activities such as social media, gaming or studying do you typically engage in on your devices?
Interviewee: I usually use SNS mostly and…and I would…in my laptop I would study, then in my phone it's for entertaining like games and videos.
Interviewer: Okay, so prior to going to bed, do you take any breaks from using your phone?
Interviewee: Um no, not really.
Interviewer: Okay *light laugh*. For your behavioral patterns, how often do you use your electronic devices at night and for how long would you estimate?

Interviewee: Umm before I fall asleep?
Interviewer: Yes, before you sleep.
Interviewee: Um I think I use usually once or three hours, but…my phone is just stick with me like- all the time.
Interviewer: Okay. Do you feel that... Wait, where am I? This is the effect of not sleeping. *laughs* I'll just cut that part out. But like, do you feel that using devices at night affects your ability to fall asleep or stay asleep?
Interviewee: Yeah, because usually I put my…I…*long pause* I play some videos or like a background music while I'm sleeping so I think that affects my sleep.
Interviewer: Okay, thank you. Have you noticed any changes in your mood, stress levels, or energy after a night of disrupted sleep?
Interviewee: Um yeah, my…temper gets shorter, and I usually get mad a lot.
Interviewer: Okay, thank you. And what emotions do you associate with sleep difficulties?such as frustration…anxiety, like words you would correlate with sleep difficulties?
Interviewee: I feel…frustrated? and I like- I feel myself that I'm such, I'm so miserable like for not sleeping early because that's uh that's not a good thing for me to do.
Interviewer: Thank you thank you that's also very true thank- thank you. Have you tried any methods to reduce your screen use at night? If yes, which ones?
Interviewee: Um I've tried to read books before going to sleep, but that didn't work well.
Interviewer: Okay. Do you use blue light filters or apps to manage screen exposure in the evening? These are like eye protection apps or features on your phone?
Interviewee: Yeah, and my screen filter is always on, so my screen is quite yellow.
Interviewer: Okay, thank you. And what other strategies have you heard about or tried to improve your sleep or mental health in relation to screen use?
Interviewee: I've tried giving my phone to my parents or to not use it or I also tried changing my phone into 2G but my mom won't allow me to do it.
Interviewer: Oh- I-I should consider those thank you I don't know I need sleep too.*laughs* Okay so we will now be moving on to the um survey questions that I had sent you the first one which is the GAD-7, the generalized anxiety disorder screener.
Interviewee: Haha I'm gonna get a high score for that, I know.
Interviewer: Yeah I will be scoring this and then telling it to you at the end of this so you can know. Yeah so for the first question over the last two weeks, how often have you been bothered by feeling nervous anxious or on edge? Zero is not at all, one is several days, two is more than half the days, three is nearly every day.
Interviewee: several days .
Interviewer: Okay, number two, over the last two weeks how often have you been feeling that you're not being able to stop or control worrying?

Interviewee: more than half the days.
Interviewer: For number three, worrying too much about different things?
Interviewee: Several days.
Interviewer: Several days…oki. For trouble relaxing over the last two weeks?
Interviewee: More than half the days.
Interviewer: Being so restless that it is hard to sit still?
Interviewee: *Light laugh* Nearly every day.
Interviewer: Becoming easily annoyed or irritated?
Interviewee: Nearly every day.
Interviewer: Feeling afraid as if something awful might happen?
Interviewee: Several days.
Interviewer: Several days…and the for if you checked off any of these problems how difficult have these problems made it for you to do work take care of things at home or get along with other people?
Interviewee: Wait again?
Interviewer: um so this one isn't uh in the one I sent you because it accidentally got cut but uh so if you checked off any of these problems, how difficult have these problems made it for you for you to do work, take care of things at home, or get along with other people socially? The first one is not difficult at all, second is somewhat difficult, third is very difficult, Fourth is extremely difficult.
Interviewee: Okay…uh somewhat.
Interviewer: Somewhat…somewhat difficult?
Interviewee: Yeah.
Interviewer: Okay. Thank you very much. I'll send you the question as well because I'm sorry that that did not send? Yeah, sorry. Okay, then we will move on to the next one, which is the Pittsburgh sleep quality index also known as the PSQI…
Interviewee: Mhm
Interviewer: Okay *clears throat* so for this one, in the instructions for it, so the following questions relate to your usual sleep habits during the past month 'only', your answers should indicate the most accurate reply for the majority of days and nights in the past month please answer all questions and during the past month when have you usually gone to bed?
Interviewee: Um…*long pause* twelve?
Interviewer: Yeah.
Interviewee: Yeah but I'm, I don't think this is accurate because whenever I sleep late, like 2, 3, or 4, and then like after the past few days, I would sleep at 6 because I uh I lack of sleep.
Interviewer: 6 a.m.? Interviewee: (same time): I don't know what to do.
Interviewer: It can be an estimate.

interviewee: No, 6 p.m.
Interviewer: Ah, 6 p.m.
Interviewee: Yeah, as soon as I go home, I would sleep and then wake up next morning.
Interviewer: Okay it can be an estimate so from 6 p.m. to what's the latest?
Interviewee: Latest? 4?
Interviewer: 4 a.m?
Interviewee: Yeah.
Interviewer: Uh then for how long has it taken you to fall asleep each night?
Interviewee: 10 to 20 minutes.
Interviewer: 10 to 20 minutes?
Interviewee: Yeah.
Interviewer: Okay. I just need to confirm because I don't know why, but my laptop's audio is like- it's going from loud to quiet…
Interviewee: Oh really?
Interviewer: Eum so if you hear me repeat I'm sorry.
Interviewee: It's okay.
Interviewer: Um when have you usually gotten up in the morning?
Interviewee: Um like- 6 a.m. yea 6 a.m. or…no, from 6 a.m. to 8.30.
Interviewer: Okay 6a.m. to 8:30…okay.
Interviewee: Yeah.
Interviewer: And then how many hours of actual sleep do you get at night this would be different from the hours you spend in bed .
Interviewee: 4 to like- 12 hours?
Interviewer: 12?
Interviewee: Yeah because whenever I don't sleep like- I would get like four hours, four hours, four hours and then 12 hours.
Interviewer: ah okay yeah I getchu this morning I woke up not in morning I woke up at 2 50 p.m.
Interviewee: Yeah it's okay.
Interviewer: Yeah happiest…happiest sleep I got in the past month. So- during the past month how often have you had trouble of sleeping because you cannot get to sleep within 10 minutes zero is not during the past month, one less than once a week, two is once or twice a week, and three is three or more times a week.
Interviewee: Less than once a week.
Interviewer: Okay. B is wake up in the middle of the night or early morning?
Interviewee: Not during the past month?
Interviewer: Yeah, during the past month….(pause due to no response and muffled noises) Pardon?

Interviewee: Yeah, during the past month.
Interviewer: Okay.
Interviewee: Not- not during the past month.
Interviewer: Not during the past month.
Interviewee: Yeah.
Interviewer: For C, it's you have to get up to use the bathroom during this past month.
Interviewee: Not during.
Interviewer: Okay.
Interviewer: The past month.
Interviewer: Cannot breathe comfortably.
Interviewee: Uh less than once a week.
Interviewee: Less than once a week…Where you cough or snore loudly?
Interviewee: …No
Interviewer: No? Okay *Light laugh*
Interviewee: Zero…Yep!
Interviewer: Where you feel too cold?
Interviewee: Ah less than once a week.
Interviewer: Feel too hot.
Interviewee: Ah less than once a week also.
Interviewer: Do you ever have bad dreams during the past month?
Interviewee: Nope! I don't even, I don't have dreams anymore. I don't know why.
Interviewee: School is taking them away for real.
Interviewee: *laughs*
Interviewer: Have you ever had pain during the past month when you try to sleep?
Interviewee: Less than once a week.
Interviewer: Gurl, you gotta get more sleep. For other reasons, please describe and then we can include um from ratings of 0, 1, 2, 3, which is still not during the past month, less than once a week, once or twice a week, three or more times a week. For your information, the other two interviewees chose their reason to be research? You can also do the same if you want.
Interviewee: Anxiety, yeah.
Interviewer: Owki,..anxiety.
Interviewee: Yep.
Interviewer: Is this three or more times a week.
Interviewee: That's- actually that's still once or twice a week.
Interviewer: Oki during the past month…did I also send you this one? okay yeah during the past month, how would you rate your sleep quality overall? very good, fairly good, fairly bad, fairly-very bad

Interviewee: fairly bad…quality?
Interviewer: yeah.
Interviewee: Oh but it's…it was fairly oh no not really fairly good, very good, very good
Interviewer: yeah?
Interviewee: I yeah quality is good, it's just the number of times.
Interviewer: Okay during the past month, how often have you taken medicine to help you sleep prescribed or over the counter?
Interviewee: What? during the past…yeah not during the past month…is it I have a question
Interviewer: Okay.
Interviewee: Is it like-... for take medicine to help you sleep does it include like- because of the pain you're having you had to take your pain reliever like paracetamol or acetaminophen to fall asleep?
Interviewer: if it's for the purpose to sleep then yes, if it's just for the purpose because you're in pain then not really
Interviewee: Mm then not during the past month.
Interviewer: Okay…Lydia don't die on me.
Interviewee: No, I just had me pain and…I don't know why.*laughs*
Interviewer: Research. School. School research. Girl, those are so many reasons possible. *Pause from typing* Okay, then the next question I'll ask you is…during the past month, how often had you had trouble staying awake while driving, eating meals, or engaging in social activities?
Interviewee: So my memory is very short term I only remember like one or two weeks only
Interviewer: That's okay, you ever have trouble with that? like staying awake at school?
Interviewee: yeah once or twice.
Interviewer: And then for the last one, it is, during the past month, how much of a problem has it been for you to keep up enthusiasm to get things done?
Interviewee: A very big problem.
Interviewer: Very big problem…Owki. Thank you very much. This concludes the interview. I will then be tallying up your score and then telling you it. Thank you very much.

Respondent 4

Interviewer: So this is my fourth interviewee. Today is October 1st, 2024. His name is ----------. So today I am currently conducting my research for our subject practical research one, qualitative research, where the research focuses on the influence of blue light exposure on sleep quality and mental health in adolescence. On this note, I have selected you to be a key informant. As such, pertinent information will be solicited through an interview through this one, where I have sent you the interview schedule for reference, which you will be asked to respond to and

accomplish by today within approximately 30 minutes. The information that you will share will greatly aid in the understanding of the state of affairs in the mentality of adolescent students, most especially from a real-world perspective considering the effects of blue students, most especially from a real-world perspective considering the effects of blue light. Rest assured that the information that you will share will be treated with utmost confidentiality and will be used only for academic purposes and classroom discussion. Respectfully, Seraphim Manuel. And I will be providing you the results of your questionnaire to you at the end of this interview. Any questions?

Interviewee: Okay will this interview be open-ended or close-ended?

Interviewer: Um hopefully close-ended with the open-end questions you will be responding with.

Interviewee: Okay.

Interviewee: Yes. So the general objectives of my qualitative research is to explore the influence of blue light exposure on sleep quality and mental health in adolescents and understand their perceptions, behaviors, and coping strategies. Some specific objectives are to assess adolescents' awareness of blue light exposure and its effects on sleep and mental health, to investigate the behavioral patterns of evening use of blue light-emitting devices among adolescents, to analyze the perceived effects of blue light exposure on sleep quality, emotional well-being, and mental health, for it to identify the coping strategies adolescents employ to mitigate the effects of blue light exposure on sleep and mental health, and then finally to recommend interventions and educational strategies to reduce the negative impacts of blue light exposure. If you need me to repeat any question in regards to connection issues or any of the such, you may do so as well as possibly if there is an issue at my end, please understand. Thank you very much. So let's hav-

Interviewee: I don't have any question regarding the things that you've mentioned. Um I just want to inquire regarding what is blue light., so I may be able to answer your questions more efficiently.

Interviewer: I apologize, but that is one of the questions for this interview.

Interviewee: …Okay

Interviewer: Yes, that is the first, actually, because this is trying to gain for the awareness of how much adolescents, specifically targeted within the year of 2008, are aware of what blue light is and what it may do.

Interviewee: Okay.

Interviewer: So for the first question I will be asking you is regards to awareness. I also sent you this if ever you wish to check. What do you know about blue light and its effects on sleep and mental health?

Interviewee: From what you've mentioned, blue light may be a type of energy released or emitted by devices.

Interviewer: Yeah.

Interviewee: And aside from that, I have no other idea regarding what exactly blue light means.
Interviewee: Okay, thank you for your honesty. Then the follow up question is, possibly have you ever heard about any influences it may have to your melatonin levels or circadian rhythms? This is in regards to your sleep schedule as well as your mental well-being.
Interviewee: Mm not really.
Interviewer: Okay, then I will be giving you some information as to what it is for you to understand what it could be. So blue light refers to a portion of the visible light spectrum with wavelengths ranging from approximately 380 to 500 nanometers. It has relatively a high energy compared to other visible light and is sometimes referred to as high energy visible HEV light. It is present in sunlight but is also emitted by artificial sources such as LED screens which is through your phones, tablets, computers, and TVs, fluorescent lighting, and some LED lights. These are normally known as your eye protection filter that your phone has…and normally the effects that they can have is that it regulates the body's circadian rhythm, this is your sleep-wake cycle, and a potential negative effect is that excessive blue light at night suppresses melatonin production, along with this ongoing research about its potential contribution to retinal damage. Though there is no confirmed evidence about long-term harm from screen use, this is particularly one of the ongoing discussions that I will be having in my interview to be concluding and discussing about. Thank you. So for the next questions, this is towards your behavioral patterns. How often do you use your electronic devices at night and for how long?
Interviewee: It depends whether I need to charge my phone, but at minimum it's probably five hours, hours?
Interviewer: Okay, minimum is five hours. What's the approximate maximum?
Interviewee: Approximate maximum would be six and if not, I can include during the daytime, probably 12 hours.
Interviewer: Okay, next question is for what type of activities such as social media, gaming, and studying do you typically engage in on these devices?
Interviewee: I…as a responsible student I engage in studying as well using my device. Most importantly right now because we are doing our research. So I need to spend more time facing the ah screen of my laptop in order to write what I must in my research and at the same time gather information from the web, however I do game time to time in order to lessen the stress, in order to keep my mental stability..at a…stable level I guess? And what was the other one that you mentioned?
Interviewer: Social media?
Interviewee: Oh yeah, of course…I use that a lot as well. So maybe less gaming, but more on studying, as of now, and a little bit of social media.
Interviewer: Okay, thank you very much. The next question is, do you take any breaks from screen use before going to bed?

Interviewee: Pardon?
Interviewer: Do you take any screen use break before going to bed?
Interviewer: Oh, screen use break. Yes. Like, do I take my time to rest my eyes from my screen? Is that what you want?
Interviewer: Yes.
Interviewee: Is that what you mean?
Interviewer: Yes…that is correct.
Interviewee: Before I sleep? No.
Interviewer: …So you go straight to bed after cutting off your phone or devices?
Interviewee: Unfortunately, I do. Because I tend to feel my eyes getting heavier after using my phone, so I immediately go to sleep after using it.
Interviewer: Okay, thank you. Then the next portion is towards perceived effects. Do you feel that using devices at night affects your ability to fall asleep or stay asleep?
Interviewee: My…Can you repeat the question?
Interviewer: Yes, do you feel that using devices at night affects your ability to fall asleep or stay asleep?
Interviewee: I think…Personally, for my case, even as much as I use my device, I would still Uh feel the exhaustion and tend to go to sleep some point. So I think my sleep schedule will still be fixed even though I…my phone is the last thing that I will touch before going to bed.
Interviewer: Okay, next one is, have you noticed any changes in your mood, stress levels, or energy after a night of disrupted sleep?
Interviewee: A night of disrupted sleep? Yes, I do. If I don't get enough sleep, I'll tend to doze off the following day. And at the same time, I'll get- be more...You get what I mean. It's like I'm not comfortable? around as much as I am when I have full sleep.
Interviewer: Yeah. Thank you very much. I understand what you mean. And what emotions do you associate with sleep difficulties? Like the term sleep difficulties what do you associate such as frustration anxiety and alike
Interviewee: it's a mix honestly but if more often than not I guess I can associate negative emotions more than positive than the positive ones yeah? like frustration worry fear, overthinking something like that
Interviewer: Okay, thank you. And then for coping strategies, have you tried any methods to reduce your screen use at night? If yes, which ones?
Interviewee: I haven't thought of that yet.
Interviewer: Yes thank you
Interviewee: Because even as of now, because even if I wanted to, I would need to use my phone and my other devices constantly due to the budget of work that I have right now as a STEM student.

Interviewer: Okay, thank you very much. Do you use blue light filters or apps to manage sreen exposure in the evening?
Interviewee: Well, not just in the evening, but within or upon using my phone, I activate the eye care or eye protection programming in order to at least minimize the radiation that is emitted by my cellular phone. And I use this all the time, not just at night but also during the day.
Interviewer: Okay, thank you.
Interviewee: -And I think this is quite comfortable for my eyes.
Interviewer: Ye ye…okay, thank you very much. What other strategies that, although you've never done any, what other strategies have you heard about or tried to improve your sleep or mental health in relation to screen use?
Interviewee: In relation to screen use…
Interviewer: Yeh
Interviewee: Maybe…I can't really think of any way that can improve my sleep schedule because if my phone will be the last thing that I'll be touching before going to bed, then most likely the reason would be because of school works. So probably if…I don't have any much to worry about regarding academics, I'd be able to go to sleep more peacefully. And also maybe, yeah-lessening my screen time as well. Because I wouldn't need to go on the web, search for different information just to report the things that I'm doing for my assignments.
Interviewer: Okay, thank you very much. So these questions were aimed to gather qualitative data on how adolescents perceive, experience, and address the impact of blue light on their daily lives while also providing insights for actionable solutions. We will now go on a temporary break before we begin on to the survey questions. I will now be pausing the survey recording.

Okay, we will now be resuming the second part of the survey. This is the Generalized Anxiety Disorder Screener or also known as the GAD-7. This is normally a psychologist's psychological test approved wherein to test the person's anxiety levels. So for over the last two weeks, how often have you been bothered by the following problems? This will be rated through these numbers of 0, 1, 2, 3, where 0 is not at all, 1 is several days, 2 is more than half the days, and 3 is nearly every day. So for over the last two weeks, how often have you been feeling nervous, anxious, or on edge?
Interviewer: From the past few days, maybe ever since I think last month, since last month. Since the start of November.
Interviewee: So is that nearly every day? More than half the days? Several days? Not at all?
Interviewee: Every day *Short laugh*
Interviewer: Okay. For number two, over the last two weeks, how often have you been not able to stop or control worrying?

Interviewee: To be honest, the only instances where I stop worrying about things is maybe when I go to sleep. Because yes, I mean, I could slack off and have fun from time to time.
But I cannot deny that there is this constant feeling within me that I need to finish something. And eventually in the following days, I will be in task to finish something inevitably.
Interviewer: Okay, so would we say that's nearly every day or more than half the days?
Interviewee: Oh three-quartzs.
Interviewer: Um, I'm sorry. It can only be rated from numbers zero to three.
Interviewee: Zero to three.
Interviewer: Yes. Three is nearly every day. Two is more than half the days.
Interviewee: Two. Okay. Number three, over the last two weeks, how often have you been worrying too much about different things?
Interviewee: How often have I been worrying about different things?
Interviewer: Yes. Once more, it's following the same format as not at all, several days, more than half the days, or nearly every day.
Interviewee: Recently?
Interviewer: Yes, within the last two weeks…
Interviewee: Three.
Interviewer: Three, Okay. And how about trouble relaxing?
Interviewee: Three.
Interviewer: Being so restless that it is hard to sit still?
Interviewee: I would still go with three.
Interviewee: Being…um becoming easily annoyed or irritated.
Interviewee: Two.
Interviewer: Afraid as if something awful might happen.
Interviewee: Two.
Interviewer: And then for the last one is, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people? Not difficult at all, somewhat difficult, very difficult, extremely difficult.
Interviewee: Extremely difficult.
Interviewer: *hiccup* Okay, and for confirmation, the symptoms began during the start of November?
Interviewee: Hmm I can notice some of the signs before November itself but they just boomed during the last month, previous month.
Interviewer: Thank you. We will then be moving to the last portion of this interview where it is the Pittsburgh Sleep Quality Index, also known as the PSQI. These are following questions that relate to your usual sleep habits during the past month ‘only’, your answer should indicate the

accurate reply for the majority of days and nights in the past month, please answer all questions, during the past month when have you gone to bed?
Interviewee: For the last month? there are nights where I go to sleep at 10…
Interviewer: Okay…
Interviewee: But most of the time, I go to sleep at 11 or later than that because my original sleep schedule should be around 9.30 to 5 a.m.
Interviewer: Okay. How long in minutes has it taken you to fall asleep each night?
Interviewee: Kind of depends on the situation, but usually it takes 5 to 10 minutes.
Interviewer: Okay. When have you usually gotten up in the morning?
Intevriewee: Excuse me?
Interviewer: What time do you usually get up in the morning?
Interviewee: …Five.
Interviewer: How many hours of sleep do you actually get at night? This may be different than the number of hours you spend in bed.
Interviewee: Six to Seven, but sometimes it's five hours.
Interviewer: Okay. Then, during the past month, how often had you had trouble sleeping because you cannot get to sleep within 30 minutes? 0 is not during the past month, 1 is less than once a week, 2 is once or twice a week, 3 is 3 or more once a week two is once or twice a week three is three or more times a week once more during the past month how often had you had trouble sleeping because you cannot get to sleep within 30 minutes?
Interviewee: …Oh wait One.
Interviewer: Huh?
Interviewee: One.
Interviewer: Oh okay thank you um during the past month how often had you had trouble to sleep because you wake up in the middle of the night or early morning?
Interviewee: No If I-...*lag*
Interviewer: Have to get up to use bathroom?
Interviewee: I mean…just one
Interviewer: Okay, cannot breathe comfortably?
Insterviewee: Cannot breathe comfortably…
Interviewer: Yes.
Interviewee: I would say…two.
Interviewer: Cough or snore loudly?
Interviewee: Cough?
Interviewer: Cough or snore loudly?
Interviewee: I don't do both.
Interviewer: Okay feel too cold?

Interviewee: Huh?...again?
Interviewer: During the past month how often had you had trouble sleeping because you feel too cold?
Interviewee: Uh one?
Interviewer: Feel too hot?
Interviewee: One.
Interviewer: Have bad dreams?
Interviewee: Okay, one.
Interviewer: You feel in pain?
Interviewee: Same.
Interviewer: Okay, so for the next question, for other reasons, please describe including how often you have had trouble sleeping because of this following the same format. For your information, two of the former respondents had decided to use their reason as research. You may do so or choose something else as the other participants had also done.
Interviewee: Can you repeat the question?
Interviewer: During the past month, how often had you had trouble sleeping due to other reasons that you may describe, including how often you have had trouble sleeping because of this reason?
Interviewee: Aside from research, I would definitely mention other curricular and extracurricular activities, especially the competition and the performance tasks that are given to us from time to us time and at the same time maybe the uh…what was the other one?
Interviewer: Uhm how often had you had trouble sleeping because of this following the same format as not during the past month less than once a week once or twice a week three or more times a week?
Interviewee: Three.
Interviewer: Okay next question is during the past month how often have you had to take medicine prescribed or over-the-counter to help you sleep?
Interviewee: Zero.
Interviewee: During the past month, how often had you had trouble staying awake while driving, eating meals, or engaging in social activity?
Interviewee: One.
Interviewer: And during the past month, how much of a problem has it been for you to keep up enthusiasm to get things done?
Interviewee: Two.
Interviewer: For the final question, during the past month, how would you rate your sleep quality overall?
Interviewee: What's the scale?

Interviewer: Very good, fairly good, fairly bad, very bad.
Interviewee: Very good, fairly bad, very bad. Very good, very bad…If I'll be honest, it's fairly bad.
Interviewer: Okay, thank you.
Interviewee: You're welcome.
Interviewer: So this concludes our interview. I may tally up all your scores and provide you the results now or send it to you through text.

Respondent 5

Interviewer: So today is the October 1st with interviewee ---------- for my practical research one for the influence of blue light exposure on sleep quality and mental health in adolescents. For your knowledge for the consent form that I had, I'm currently conducting my research for our subject practical research one qualitative research where the research focus is on the influence of blue light exposure on sleep quality and mental health in adolescents. On this note, I have selected you to be a key informant. As such, pertinent information will be solicited through an interview, as well as the attached interview schedule for reference, which I have sent you, which you will be asked to respond to within approximately less than 30 minutes. The information that you share will be greatly aided in the understanding of the state of affairs in the mentality of adolescent students, most especially from a real-world perspective, considering the effects of blue light. Rest assured that the information that you will share will be treated with utmost confidentiality and will be used only for academic purposes and classroom discussion. For the general objective, this is to explore the influence of blue light exposure on sleep quality and mental health in adolescents and understand their perceptions and behaviors and coping strategies. For specific objectives, this is to assess adolescents' awareness of blue light exposure and its effects on sleep and mental health, to investigate the behavioral patterns of evening use of blue light emitting devices among adolescents, to analyze the perceived effects of blue light exposure on sleep quality, emotional well-being, and mental health, fourth, to identify the coping strategies adolescents employ to mitigate the effects of blue light exposure on sleep and mental health, as well as to recommend interventions and educational strategies to reduce the negative impacts of blue light exposure. We will now begin the open-ended questions if you are ready.
Interviewee: Yep
Interviewer: Okay, so for the first question, this portion goes to awareness. What do you know about blue light and its effects on sleep and mental health?
Interviewee: Blue light as itself is like on what I know is a visible light spectrum. It is emitted by the sun or artificially like the LED lights or what we have on our computer screens and smartphone screens and well on what I know on sleep and mental health I think it does influence well in the brain function I guess.

Interviewer: Okay thank you and have you ever heard about how it influences melatonin levels or Circadian rhythms? this is your sleep-wake cycle.
Interviewee: The topic of suppression of melatonin or influence of melatonin you mean?
Interviewer: Yes that's correct.
Interviewee: So in the term of blue light exposure I think especially during the evening inhibits like the secretion of melatonin like the hormones that are responsible for regulating sleep I guess this makes it harder to fall asleep and it can disturb the sleep quality.
Interviewer: Thank you so much. For the next question, this is towards behavioral patterns. How often do you use electronic devices at night and for how long?
Interviewee: For how long do I use my electronics in the night? I think I'd have to check on the tool if that's okay…
Interviewer: That's okay with me, Thank you.
Interviewee: So at night, it says in my activity details in the screen time limits,
it says that I use it approximately 4 hours in night.
Interviewer: Okay. Is this average?
Interviewee: For nights only?
Interviewer: Yes.
Interviewee: Or for the whole day? If it's 4 nights, average is 4 hours and 37 minutes.
Interviewee: Okay, thank you so much. We will then move to the next question. As to what type of activities such as social media, gaming, or studying do you typically engage in on your devices?
Interviewee: I usually do games since I don't really check on social media much ‘cause I think social media impacts well your sleep behavior more than usually doing your daily games with what they call this? with having the daily quest so I guess I take…little more time on gaming then instead of socialize or the social media things.
Interviewer: That's okay. Thank you. And do you take any breaks from screen use prior to going to bed?
Interviewee: Prior to going to bed? Yeah, I take screen breaks because usually I take a shower before going to bed.
Interviewer: Thank you. For the next question, this is the perceived effects, do you feel that using devices at night affects your ability to fall asleep or stay asleep?
Interviewee: Well, I have insomnia either way, without my phone or with it, so I think it doesn't really affect my sleeping ability.
Interviewer: Okay. Have you noticed any changes in your mood, stress levels, or energy after a night of disrupted sleep, if you ever have such?
Interviewee: Disrupted sleep…In disruptive sleep you mean like what exactly?

Interviewer: Um disrupted sleep as in you have woken up due to your devices or external factors..
Interviewee: External factors meaning only devices meaning uh if it was devices no since I always have my notification off.
Interviewer: Okay. However, if you did have a disrupted sleep from other possible factors, would you say that your mood, stress levels, or energy are different?
Interviewee: Other factors?
Interviewer: Yeah.
Interviewee: Other factors, if it was for chores, being woken up in the night for chores, I guess so, that changes my mood.
Interviewer: Okay, moving on, what emotions do you associate with the term of sleep difficulties? These could be frustration or anxiety or such the like.
Interviewee: The emotion associated with sleep difficulties, I guess, frustrations and, you know, everyone wants to have like their daily sleep. But as you know, we have so much things to do as a grade 11 student being bombarded with so much stuff to do so I guess yeah um that's one of the factors of my emotions and getting difficulty of sleep yeah frustration definitely.
Interviewee: Thank you so much we will then be moving to the coping strategies if you have any methods that you have tried to reduce your screen use at night if yes um which ones?
Interviewee: I don't really reduce my screen time since I think four hours isn't that much. Four hours and 37 minutes, I mean, or specific, isn't that much. And if I would, I think I'd keep my phone as far away, I guess.
Interviewer: Okay, thank you. Do you use blue light filters or apps to manage screen exposure in the evening? These are also known as eye protection or the warm light that your devices emit.
Interviewee: No, I don't use that at all.
Interviewer: Okay. What other strategies have you heard about or tried possibly to improve your sleep or mental health in relation to screen use?
Interviewee: What are the strategies? Specifically in screen use, I have none because I've never really thought of it as a problem.
Interviewer: Okay, thank you so much. These questions aimed to gather qualitative data on how adolescents perceive, experience, and address the impact of blue light on their daily lives, while also providing insights for actionable solutions. We will now be moving on to the next portion, unless you wish to have a momentary break.
Interviewee: Nope, it's all good.
Interviewer: Okay, so we will be moving on to the next portion. This is a psychological test, also known as the Generalized Anxiety Disorder Screener, known as the GAD-7. This is to test over your anxiety from the last two weeks, if this is alright with you.
Interviewee: Yep.
Interviewer: So, over the last two weeks, how often have you been feeling nervous, anxious, or on edge? To rate this, it's from 0, 1, to 3, where 0 is not at all, 1 is several days, 2 is more than half the days, and 3 is nearly every day.

Interviewee: 1, Several days.
Interviewer: Okay. Over the last two weeks, how often have you been bothered by not being able to stop or control worrying?
Interviewee: 0, Not at all.
Interviewer: Number 3 is worrying too much about different things.
Interviewee: One, several days.
Interviewer: Have you had trouble relaxing over the past two weeks?
Interviewee: Zero, not at all.
Interviewer: Being so restless that it is hard to sit still?
Interviewee: Zero, not at all.
Interviewer: Becoming easily annoyed or irritated?
Interviewee: One, several days.
Interviewer: And then feeling afraid as if something awful might happen?
Interviewee: Zero, not at all.
Interviewer: Then for the final one for this survey, if you um checked off any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people? Is it not difficult at all, somewhat difficult, very difficult, or extremely difficult? Interviewee: Could you make that more specific?
Interviewer: Okay, so the problems that you did check was how you felt nervous, anxious, on edge, you worry about different things, and you become easily annoyed or irritated.
How difficult have these problems made it for you to do your work, take care of things at home, or get along with people?
Interviewee: It's somewhat difficult.
Interviewer: And when do you believe that these symptoms began?
Interviewee: The symptoms began when well there were a lot of things that were given to us but then again I don't really um well I allocate time for it and with the allocation of my time for it mostly being night um which means that since I work better at night having insomnia and all I uh guess it just is somewhat difficult when I try to do it in the morning. So that is why my moods may be irritable in the morning rather than, you know, being focused on what I'm doing since I see that I work better at night.
Interviewer: Okay, thank you very much. I will be at the end of this interview telling you your results, if that's okay with you. And then for the last portion, you will be taking the Pittsburgh Sleep Quality Index also known as the PSQI so the following questions relate to your usual sleep habits during the past month 'only', your answers should indicate the most accurate reply for the majority of days and nights in the past month giving an average response or approximate response is also possible and yes we'll now begin so during the past month when have you usually gone to bed at night?

Interviewee: Um it will be about uh let's say if it was this past month meaning that we had well uh pr1 yes I'd say that it was 11.
Interviewer: Okay.
Interviewee: Exactly 11.
Interviewer: Okay How long in minutes has it taken you to fall asleep each night?
Interviewee: Each night, I'd say two hours.
Interviewer: Pardon?
Interviewee: Oh, minutes. Well, mine's not really minutes. I take a lot of time to procrastinate to sleep.
Interviewer: O-oh Okay. That's how many hours?
Interviewee: Two.
Interviewer: Okay…Okay, yes. Thank you very much for your honesty. Um when have you usually gotten up in the morning?
Interviewee: In the morning, school time. If it was, well, the weekends. Let's say...
Interviewer: The majority average.
Interviewee: Majority?
Interviewer: Yes.
Interviewee: Well, I wake up at 7.
Interviewer: Okay. And then how many hours do you actually get to sleep at night? This may be different from the number of hours you spend in bed.
Interviewee: *Long pause* Nine hours.
Interviewer: Okay. We will then be moving to the next questions, which may be answered with 0, 1, 2, 3, where 0 is not during the past month, 1 is less than once a week, 2 is once or twice a week, and 3 is three or more times a week. So during the past month, how often have you had trouble sleeping because you cannot get to sleep within 30 minutes?
Interviewee: Three or more weeks time only.
Interviewer: Okay, thank you. When you wake up in the middle of the night or early morning?
Interviewee: Not during the past months.
Interviewer: Okay. Have to get up to use the restroom?
Interviewee: Not during the past month.
Interviewer: Cannot breathe comfortably?
Interviewee: Not during the past month.
Interviewer: Cough or snore loudly?
Interviewee: Not during the past month.
Interviewer: Feel too cold?
Interviewee: Not during the past month.
Interviewer: Feel too hot?

Interviewee: Not during the past month.
Interviewer: Have bad dreams?
Interviewee: Once or twice a week.
Interviewer: Okay. Have pain?
Interviewee: Not during the past month.
Interviewer: For other reasons, please describe including how often you have had trouble sleeping because of these reasons. former interviewees had decided to use research, anxiety, and insomnia as their reasons?
Interviewee: Well, with all the questions say…like, that's the reason?
Interviewer: Yes. Not really, but for this one here, where you can have the opportunity to make your own separate reason as to why you have trouble sleeping the past month.
Interviewee: Well, I've always had trouble sleeping, but I guess during this time it's more on yeah research PR1.
Interviewer: Okay and how long would you say that be from the range of 0 to 3?
Interviewee: Not always but since it's just the past month yeah three or more times a week.
Interviewer: Okay and during the past month how often have you taken medicine prescribed or over-the-counter to help you sleep?
Interviewee: Oh how would I rate my sleep quality?
Interviewer: Um during the past month how often have you taken medicine prescribed or over-the-counter to help you sleep? Still once more with the not during the past month to three or more times a week ranking.
Interviewee: Okay, not during the past month.
Interviewer: Okay thank you. During the past month how often have you had trouble staying awake while driving, eating meals, or engaging in social activity?
Interviewee: …Less than once a week.
Interviewer: Less than once a week…Okay. During the past month, how much of a problem has it been for you to keep up enthusiasm to get things done?
Interviewee: Not during this month.
Interviewer: Okay, thank you. And for the final question, during the past month, how would you rate your sleep quality overall? Very good, fairly good, fairly bad, very bad?
Interviewee: It's fairly good.
Interviewer: Okay this then concludes the interview I will be giving you results soon after thank you very much.

Coding framework

Thematic Framework

Interviewee	Excerpt	Code	Theme
Respondent 1	"It's often caused from procrastinating, especially in research."	procrastination, stress	Sleep disruption and stress
Respondent 2	"When you have a lack of sleep, you're more prone to stress or a headache."	lack of sleep, stress	Sleep disruption and health
Respondent 3	"My temper gets shorter, and I usually get mad a lot."	mood, frustration	Emotional impact of sleep issues
Respondent 4	"I get more... not comfortable around as much as I am when I have full sleep."	comfort, disrupted sleep	Sleep quality and well-being
Respondent 5	"Frustration definitely... we have so much to do as a grade 11 student."	frustration, workload	Workload and stress

Content Framework

Interviewee	Excerpt	Code	Theme
Respondent 1	"Normally, I'd go for like 6 hours or up to 14 hours if it's weekends or summer break."	device usage, frequency	Behavioral patterns
Respondent 2	"I use my phone every night along with my computer."	frequency, multitasking	Behavioral patterns
Respondent 3	"I usually use SNS mostly, and on my laptop I study, then on my phone, it's for games."	activity types	Device usage patterns
Respondent 4	"Most importantly right now because we are doing our research."	academic use	Purposeful device usage
Respondent 5	"I take screen breaks because usually I take a shower before going to bed."	screen break	Coping strategies

Narrative Framework

Interviewee	Excerpt	Code	Theme
Respondent 1	"Using devices at night often leads me to overthink until I fall asleep around 4 or 5."	key event, emotional	Sleep disruption
Respondent 2	"The best way to avoid blue light is just to stop using gadgets in general."	turning point, solution	Awareness and strategies
Respondent 3	"I feel myself that I'm so miserable for not sleeping early."	emotional, realization	Emotional impact
Respondent 4	"I think my sleep schedule will still be fixed even though I use my phone."	reflection, habitual	Behavioral consistency
Respondent 5	"I work better at night, having insomnia and all."	key event, adaptation	Coping with challenges

Living Stone International School, Inc.
(A Non-stock, non-profit Educational Institution)
43-B Marlim Blvd. Extension Corner Debra Street, Pulung Maragul
Angeles City. The Philippines 2009
Telephone No. (045) 885 2596, (045) 436 2704
Jesus first. Others second. Yourself third.

Sample accomplished survey tool

GAD-7 Anxiety

Over the last two weeks, how often have you been bothered by the following problems?	Not at all	Several days	More than half the days	Nearly every day
1. Feeling nervous, anxious, or on edge	0	1	2	3
2. Not being able to stop or control worrying	0	1	2	3
3. Worrying too much about different things	0	1	2	3
4. Trouble relaxing	0	1	2	3
5. Being so restless that it is hard to sit still	0	1	2	3
6. Becoming easily annoyed or irritable	0	1	2	3
7. Feeling afraid, as if something awful might happen	0	1	2	3

Column totals _____ + _____ + _____ + _____ =

Total score _____

If you checked any problems, how difficult have they made it for you to do your work, take care of things at home, or get along with other people?

Not difficult at all Somewhat difficult Very difficult Extremely difficult ☐ ☐ ☐ ☐

Source: Primary Care Evaluation of Mental Disorders Patient Health Questionnaire (PRIME-MD-PHQ). The PHQ was developed by Drs. Robert L. Spitzer, Janet B.W. Williams, Kurt Kroenke, and colleagues. For research information, contact Dr. Spitzer at ris8@columbia.edu. PRIME-MD® is a trademark of Pfizer Inc. Copyright© 1999 Pfizer Inc. All rights reserved. Reproduced with permission

Scoring GAD-7 Anxiety Severity

This is calculated by assigning scores of 0, 1, 2, and 3 to the response categories, respectively, of "not at all," "several days," "more than half the days," and "nearly every day." GAD-7 total score for the seven items ranges from 0 to 21.

0–4: minimal anxiety 5–9: mild anxiety 10–14: moderate anxiety 15–21: severe anxiety

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Name:_____ Date: _____

Pittsburgh Sleep Quality Index (PSQI)

Instructions: The following questions relate to your usual sleep habits during the past month only. Your answers should indicate the most accurate reply for the majority of days and nights in the past month. **Please answer all questions.**

1. During the past month, what time have you usually gone to bed at night? _____
2. During the past month, how long (in minutes) has it usually taken you to fall asleep each night? _____
3. During the past month, what time have you usually gotten up in the morning? _____
4. During the past month, how many hours of actual sleep did you get at night? (This may be different than the number of hours you spent in bed.) _____

5. During the past month, how often have you had trouble sleeping because you…	Not during the past month	Less than once a week	Once or twice a week	Three or more times a week
a. Cannot get to sleep within 30 minutes				
b. Wake up in the middle of the night or early morning				
c. Have to get up to use the bathroom				
d. Cannot breathe comfortably				
e. Cough or snore loudly				
f. Feel too cold				
g. Feel too hot				
h. Have bad dreams				
i. Have pain				
j. Other reason(s), please describe:				
6. During the past month, how often have you taken medicine to help you sleep (prescribed or "over the counter")?				
7. During the past month, how often have you had trouble staying awake while driving, eating meals, or engaging in social activity?				
	No problem at all	Only a very slight problem	Somewhat of a problem	A very big problem
8. During the past month, how much of a problem has it been for you to keep up enough enthusiasm to get things done?				
	Very good	Fairly good	Fairly bad	Very bad
9. During the past month, how would you rate your sleep quality overall?				

	No bed partner or room mate	Partner/room mate in other room	Partner in same room but not same bed	Partner in same bed
10. Do you have a bed partner or room mate?				
	Not during the past month	Less than once a week	Once or twice a week	Three or more times a week
If you have a room mate or bed partner, ask him/her how often in the past month you have had:				
a. Loud snoring				
b. Long pauses between breaths while asleep				
c. Legs twitching or jerking while you sleep				
d. Episodes of disorientation or confusion during sleep				
e. Other restlessness while you sleep, please describe:				

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Scoring the PSQI

The order of the PSQI items has been modified from the original order in order to fit the first 9 items (which are the only items that contribute to the total score) on a single page. Item 10, which is the second page of the scale, does not contribute to the PSQI score.

In scoring the PSQI, seven component scores are derived, each scored 0 (no difficulty) to 3 (severe difficulty). The component scores are summed to produce a global score (range 0 to 21). Higher scores indicate worse sleep quality.

Component 1: Subjective sleep quality—question 9

Response to Q9	Component 1 score
Very good	0
Fairly good	1
Fairly bad	2
Very bad	3

Component 1 score:_____

Component 2: Sleep latency—questions 2 and 5a

Response to Q2	Component 2/Q2 subscore
≤ 15 minutes	0
16-30 minutes	1
31-60 minutes	2
> 60 minutes	3

Response to Q5a	Component 2/Q5a subscore
Not during past month	0
Less than once a week	1
Once or twice a week	2
Three or more times a week	3

Sum of Q2 and Q5a subscores	Component 2 score
0	0
1-2	1
3-4	2
5-6	3

Component 2 score:_____

Component 3: Sleep duration—question 4

Response to Q4	Component 3 score
> 7 hours	0
6-7 hours	1
5-6 hours	2
< 5 hours	3

Component 3 score:_____

Component 4: Sleep efficiency—questions 1, 3, and 4

Sleep efficiency = (# hours slept/# hours in bed) X 100%

hours slept—question 4

hours in bed—calculated from responses to questions 1 and 3

Sleep efficiency	Component 4 score
> 85%	0
75-84%	1
65-74%	2
< 65%	3

Component 4 score:_____

Component 5: Sleep disturbance—questions 5b-5j

Questions 5b to 5j should be scored as follows:

Not during past month	0
Less than once a week	1
Once or twice a week	2
Three or more times a week	3

Sum of 5b to 5j scores	Component 5 score
0	0
1-9	1
10-18	2
19-27	3

Component 5 score:_____

Component 6: Use of sleep medication—question 6

Response to Q6	Component 6 score
Not during past month	0
Less than once a week	1
Once or twice a week	2
Three or more times a week	3

Component 6 score:_____

Component 7: Daytime dysfunction—questions 7 and 8

Response to Q7	Component 7/Q7 subscore
Not during past month	0
Less than once a week	1
Once or twice a week	2
Three or more times a week	3

Response to Q8	Component 7/Q8 subscore
No problem at all	0
Only a very slight problem	1
Somewhat of a problem	2
A very big problem	3

Sum of Q7 and Q8 subscores	Component 7 score
0	0
1-2	1
3-4	2
5-6	3

Component 7 score:_____

Global PSQI Score: Sum of seven component scores:_____

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Citation: Buysse, DJ, Reynolds CF, Monk TH, Berman SR, Kupfer DJ: The Pittsburgh Sleep Quality Index (PSQI): A new instrument for psychiatric research and practice. Psychiatry Research 28:193-213, 1989

Interview schedule

"The Influence of Blue Light Exposure on Sleep Quality and Mental Health in STEM Students Ranging ages 15 to 16"

General Objective

To explore the influence of blue light exposure on sleep quality and mental health in adolescents and understand their perceptions, behaviors, and coping strategies.

Specific Objectives

1. To assess adolescents' awareness of blue light exposure and its effects on sleep and mental health.
2. To investigate the behavioral patterns of evening use of blue light-emitting devices among adolescents.
3. To analyze the perceived effects of blue light exposure on sleep quality, emotional well-being, and mental health.
4. To identify the coping strategies adolescents employ to mitigate the effects of blue light exposure on sleep and mental health.
5. To recommend interventions and educational strategies to reduce the negative impacts of blue light exposure.

Awareness

- What do you know about blue light and its effects on sleep and mental health?
- Have you ever heard about how blue light influences melatonin levels or circadian rhythms?

Behavioral Patterns

- How often do you use electronic devices at night, and for how long?
- What types of activities (e.g., social media, gaming, studying) do you typically engage in on these devices?
- Do you take any breaks from screen use before going to bed?

Perceived Effects

- Do you feel that using devices at night affects your ability to fall asleep or stay asleep?
- Have you noticed any changes in your mood, stress levels, or energy after a night of disrupted sleep?
- What emotions do you associate with sleep difficulties (e.g., frustration, anxiety, etc.)?

Coping Strategies

- Have you tried any methods to reduce your screen use at night? If yes, which ones?
- Do you use blue light filters or apps to manage screen exposure in the evening?
- What other strategies have you heard about or tried to improve your sleep or mental health in relation to screen use?

These questions aim to gather qualitative data on how adolescents perceive, experience, and address the impact of blue light on their daily lives, while also providing insights for actionable solutions.

Signed consent form

PURPOSE OF THE STUDY

You are invited to participate in a research study titled "The Influence of Blue Light Exposure on Sleep Quality and Mental Health in STEM Ranging ages 15 to 16". This study aims to understand the effects of blue light from electronic devices on sleep and mental health among adolescents aged 15 to 16. Your participation will contribute to identifying patterns, experiences, and coping mechanisms that can inform educational and technological interventions.

STUDY PROCEDURES

Participation in this study involves:

- *Completing a survey with open-ended and Likert-scale questions about your screen use and its effects.*
- *Participating in a semi-structured interview to share personal experiences and insights.*
  The interview may include questions about your sleep patterns, emotional state, and strategies for managing device use at night.

DURATION

- *The survey will take approximately 15–20 minutes to complete.*
- *The interview session will last 30–45 minutes.*
  Your involvement in this research concludes after these activities unless you are part of follow-up data verification.

VOLUNTARY PARTICIPATION

Your participation in this study is entirely voluntary. You can decline to participate or withdraw at any time without any penalty or effect on your relationship with the researcher or institution. If you choose to cancel, your data will be securely deleted.

RISKS

There are minimal risks associated with this study. Some questions may address sensitive topics, such as emotional well-being, which could cause temporary discomfort. You are free to skip any question or end your participation at any point.

BENEFITS

There may be no direct benefits to your participation. However, your involvement will help advance understanding of adolescent health in the digital age, potentially guiding interventions to improve sleep hygiene and mental health outcomes.

CONFIDENTIALITY

Your responses will remain anonymous, and every effort will be made to maintain confidentiality:

- *Your identity will be replaced with a code or pseudonym in all notes and documents.*
- *All data will be securely stored in password-protected files and/or locked cabinets accessible only to the researchers.*
- *Audio recordings, if any, will be deleted after transcription and analysis.*

CONTACT INFORMATION

This study was approved by the Research Ethics Review Committee of LSIS. If you have any questions at any time about this study, or if you experience any non-normative sensations as a result of participation, you may contact the researcher whose contact information is on the first page. If you have any questions regarding your rights as a research participant, or if problems arise that you do not feel you can discuss with the researchers, please feel free to contact rsierra.faculty@livingstone.ph.education

CONSENT

I have read the provided information, or it has been read to me. I have had the opportunity to ask questions about it and any questions I have been asked have been answered to my satisfaction. I understand that I will be given a copy of this form, and the researcher will keep another copy on file. I consent voluntarily to be a participant in this study.

Print Name of Participant_____
Signature of Participant _____
Date 12_/_12__/_2024__
Day/month/year

Print Name of Researcher Seraphim Maneul_
Signature of Researcher _____
Date 12_/_12__/_2024__
Day/month/year

Print Name of Impartial Witness___N/A_____
Signature of Impartial Witness ___N/A_____
Date 12_/_12__/_2024__
Day/month/year

Powerpoint presentation (appendix)

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The Influence of Blue Light Exposure on Sleep Quality & Mental Health in LSIS STEM Students Aged 15-16

Manuel Seraphim 11 - STEM

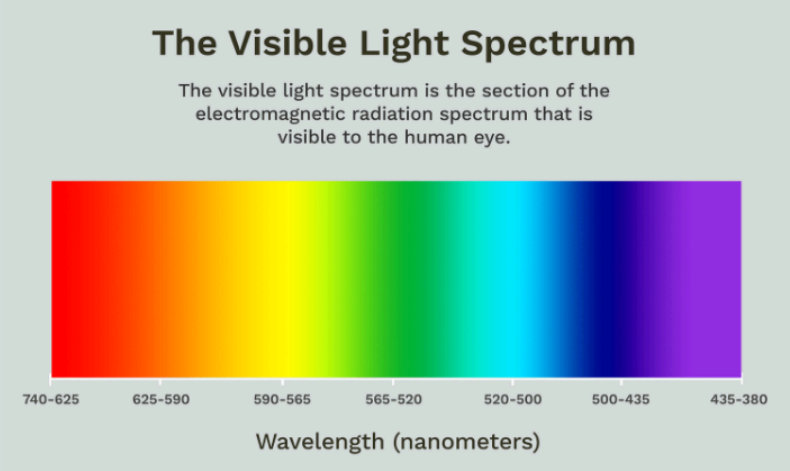
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Background

- Increasing reliance on electronic devices among adolescents
- Blue light affects melatonin and Circadian rhythms
- Importance of studying STEM students aged 15-16

Background

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Problem Statement

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- Pervasive evening use disrupts sleep & mental health
- Limited research on tailored interventions for adolescents

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Objectives

- Explore adolescents' awareness of Blue Light exposure
- Investigate evening use behaviors
- Identify coping strategies & recommend interventions

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RRL: Overview

- Key Findings: Blue Light suppresses melatonin, disrupts circadian rhythms
- Increased risks of anxiety, depression, and fatigue

Supporting Evidence

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- Evidence linking screen time to poor sleep quality & mental health issues
- Effective interventions include Blue Light filters and Behavior modifications

Conclusion

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Emphasizes the need for awareness and strategies to mitigate effects

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Participants

- STEM students aged 15-16 of Livingstone International School
- Selected via purposive sampling

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Data Collection

- Surveys: Pittsburgh Sleep Quality Index(PSQI), General Anxiety Disorder-7(GAD-7)
- Semi-structured interviews for qualitative insights

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Ethical Considerations

Informed consent, confidentiality, and voluntary participation

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Awareness

- Moderate knowledge
- Misconceptions about Blue Light Effects

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Behavioral Patterns

Frequent evening screen use for academics, social media, and gaming

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Perceived Effects

Poor sleep quality linked to

- Irritability
- Fatigue
- Focus issues

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Strengths & Limitations

Strengths

- Focus on STEM students

Limitations

- Context-specific results, small sample size

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Future Research

Diverse demographics and long-term studies needed

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Summary

- Blue Light Exposure significantly impacts sleep quality and mental health
- Need for actionable solutions educational interventions

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Call to Action

Stakeholders:

- Schools
- Parents
- Technology designers

Emphasis on fostering healthier digital habits